

CONTROLLED FLIGHT AND AERODYNAMIC STUDY: MIT OPENROCKET AND RASAERO II

MIT ROCKET TEAM — FLIGHT SIMULATION STUDIES

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MIT OPENROCKET
RASAERO II

Five flight designs — two motors
92 aerodynamic configurations
Windows 11 virtual machine

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Purpose of this document. This report compares ten paired flight predictions for five newly constructed OpenRocket models. It separates export and model-description differences from aerodynamic and numerical differences, records the controlled Windows execution, and identifies evidence-based improvements to MIT OpenRocket. Eight cases have matched geometry and roughness; the two rough-finish cases are retained as an explicitly confounded export diagnostic. A follow-on investigation isolates aerodynamic mechanisms with an initial 25-case investigation and a 92-configuration coefficient atlas, without modifying MIT OR. Agreement between programs is not physical validation against measured flight data.

1 Principal findings

The main disagreement is aerodynamic drag, with a strong dependence on fin section. After matching geometry and flow settings, RASAero predicts 1.0-27.8% lower apogee in the eight clean cases. Heavy beveled designs agree within 1.61%; the largest gaps occur on the slender rockets with square-edged fins.

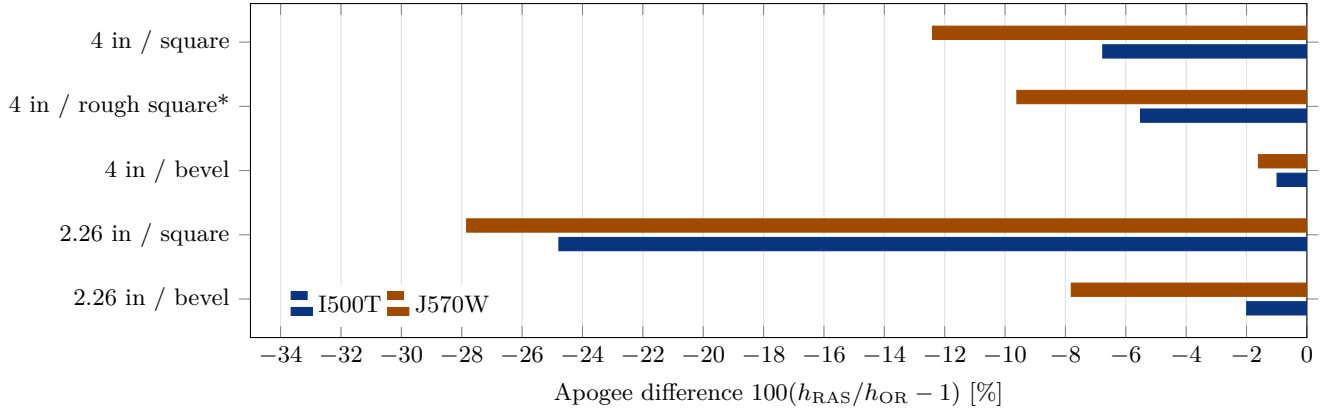


Figure 1. Apogee differences after matching geometry and fully turbulent flow. B02 (asterisk) retains a numerical roughness mismatch and is excluded from clean solver attribution.

* The rough-finish design B02 has unequal numerical roughness and is excluded from clean simulation attribution. Negative values mean RASAero predicts a lower apogee. All primary RASAero runs have All Turbulent Flow enabled.

1.1 What the study establishes

Replacing only MIT OR's drag coefficient with the curve recorded by RASAero reduces the eight clean apogee differences to **0.04-0.86%**. This supports drag modeling as the dominant cause. It does not determine which program better predicts real flight; no flight measurements were used.

Three export issues need attention: turbulent-flow settings, beveled-fin geometry, and physical roughness mapping. A separate MIT time-step override is confirmed, but actual half-step refinement changes apogee by less than 0.02% here.

Aerodynamic follow-up. Section 10 separates body and net fin contributions. It identifies a near-zero tangent-ogive nose-pressure term, loss of bevel sensitivity in the detached-shock fallback, and a body-length dependence in fin friction. An independent nose-term trial reduces mean absolute body-drag error by 84% over six configurations at Mach 1.32-3. No MIT OR application/source changes were made.

RASAero was configured, run and exported through the Windows GUI. MIT OR used the installed MIT application JAR through its simulation/export APIs; all ten baseline cases and ten requested-step repeats were also run inside the VM. Mac/Windows apogees match exactly. This is a flight-prediction study, not a computational-speed benchmark.

Completed extension. Section 11 expands the data to 92 configurations and 82,800 common Mach/angle points, with 42 plot sheets. It tests nose shapes, fin geometry, physical roughness, transition, nozzle area, scale, normal force and CP without modifying the installed OR engine.

2 Experimental design

The rockets deliberately remove appendages, active control and multi-stage complications. B01/B03 and B04/B05 isolate fin-edge treatment at fixed planform and mass. B02 exposes the roughness mapping problem. These are simulation fixtures, not qualified flight hardware designs.

Table 1. Nominal geometry and dry mass properties.

Parameter	B01 / B02 / B03	B04 / B05
Body diameter	4.000 in (101.60 mm)	2.260 in (57.404 mm)
Overall length	60.000 in (1.524 m)	50.000 in (1.270 m)
Tangent-ogive nose / body	12 / 48 in	10 / 40 in
Fins: count; root / tip chord	4; 6 / 2 in	3; 5 / 2 in
Fin span / leading-edge sweep	3.25 / 4 in	2.5 / 2.5 in
Thickness; aft-edge inset	0.125 in; 0.500 in	0.125 in; 0.500 in
Dry structure mass / dry CG	6.000 kg / 33 in from tip	1.000 kg / 27 in from tip

Table 2. Design matrix and fin-edge treatment.

Design	Surface finish	Fin section
B01 smooth square	Zero roughness	Square
B02 rough square *	OR 60 micrometers; RAS 30.48 micrometers	Square
B03 smooth bevel	Zero roughness	0.25 in nose bevel; square trailing edge
B04 slender square	Zero roughness	Square
B05 slender bevel	Zero roughness	0.25 in nose bevel; square trailing edge

2.1 Shared flight conditions

AeroTech I500T-14A and J570W use the same numerical RASP motor entries, preserved in models/study-motors.eng. Motors are aft-flush with zero overhang and no ignition delay. Dry mass/CG overrides include the structure and recovery system; motor mass and CG are added separately.

Launch altitude 0 m; vertical 2 m rail; zero wind and turbulence; 15 C and 101325 Pa at sea level. MIT OR uses ISA, latitude 45 degrees, longitude 0 and flat-earth geometry. RASAero receives 59 F, 29.9214 inHg and a 6.5617 ft rail. Both use a 24 in parachute, Cd 0.8, at apogee with no delay. Motor ejection is disabled in OR.

No lugs, rail guides, camera pods, tabs, fillets, transitions, fin cant, drag overrides or controllers. RASAero nozzle diameter is zero in these flight runs; later aerodynamic-only probes vary it to examine power-on/off drag sensitivity. Primary runs match the fully turbulent assumption: OR perfectFinish=false; RAS All Turbulent Flow=true.

3 Model equivalence and remaining differences

Table 3. Model-equivalence audit and remaining implementation differences.

Item	Observed difference / treatment	Consequence
Fin-section export	MIT TRIANGULAR exports as Square with a warning. In the GUI, B03/B05 were changed to Hexagonal Blunt Base, FX1=0.25 in, LE radius=0.	Primary bevel runs preserve the pointed leading bevel and blunt trailing edge. Raw exports are retained.
Flow assumption	Exporter writes Turbulence=False; MIT OR uses perfectFinish=false. All Turbulent Flow was enabled in RAS for every primary case.	A settings mismatch is removed before solver attribution. Default-flow runs are retained as sensitivity data.
Surface roughness	B02: OR NORMAL=60 micrometers; RAS Rough Camouflage Paint=0.0012 in=30.48 micrometers. All other designs use zero.	B02 is a mapping diagnostic, not an equivalent-model comparison.
Geometry and units	RAS body length is rounded down by 0.0001 in. Other checked external dimensions match the export.	2.54 micrometer length rounding is negligible at the displayed precision.
Mass and CG	OR retains components and motors; RAS uses entered launch mass/CG and motor data. Launch errors <0.023 g and <0.003 mm.	Initial mass properties match closely. Exported burn mass histories still differ by up to 2.28 g / 6.00 g for I500 / J570.
Dynamics	RAS no-wind mode is 2DOF; MIT OR retains its dynamics solver. RAS ascent angle of attack is exactly zero.	This axial benchmark does not compare wind response, damping, CP accuracy or control dynamics.
Atmosphere / gravity	ISA versus RAS U.S. Standard Atmosphere implementation; equal sea-level inputs. OR explicitly uses flat geometry.	Implementations and numerical evaluation remain possible sources of sub-percent residuals.
Sampling / integration	RAS CSV requested every 0.01 s; OR actual step is about 0.0025 s. RAS Mach, velocity and force columns show sample staggering.	Use direct exported peaks and explicitly matched Mach for aerodynamic comparisons; avoid inferring atmosphere from row-wise force balance.
Recovery / internals	Same nominal chute diameter, Cd and apogee trigger. Construction/inertia details are aggregated in RAS.	Ascent metrics are primary. Descent and landing loads are not validated.

RASAero was run with a per-process English number-format launcher to avoid the VM locale misreading decimal points. The installed application binary and global Windows locale were not modified. The saved native files and CSV outputs were checked for valid decimal values.

4 Flight statistics

Primary results: fully turbulent in both programs. Heights are above launch level (also MSL here). Maximum speed and Mach are ascent values. Percent difference is $100 \times (\text{RAS} / \text{OR} - 1)$. I500 denotes the exact I500T-14A curve.

$$\delta_h = 100 \left(\frac{h_{\text{RAS}}}{h_{\text{OR}}} - 1 \right) \%.$$
 (1)

Table 4. Primary ascent performance. Percent differences use MIT OR as denominator.

Case	Motor	Apogee OR / RAS (m)	Difference	Max speed OR / RAS (m/s)	Max Mach OR / RAS
B01	I500T	376.2 / 350.8	-6.76%	82.4 / 81.7	0.242 / 0.240
B01	J570W	889.6 / 779.3	-12.40%	131.7 / 128.5	0.388 / 0.378
B02 *	I500T	369.7 / 349.3	-5.51%	82.2 / 81.6	0.242 / 0.240
B02 *	J570W	851.8 / 769.9	-9.61%	130.6 / 128.1	0.385 / 0.377
B03	I500T	380.4 / 376.6	-0.99%	82.6 / 82.5	0.243 / 0.242
B03	J570W	910.9 / 896.3	-1.61%	132.3 / 131.9	0.389 / 0.388
B04	I500T	2027.4 / 1525.0	-24.78%	364.5 / 349.7	1.074 / 1.030
B04	J570W	2811.8 / 2029.1	-27.84%	479.6 / 437.4	1.417 / 1.291
B05	I500T	2252.6 / 2207.7	-1.99%	369.6 / 368.5	1.089 / 1.086
B05	J570W	3076.2 / 2836.0	-7.81%	486.1 / 463.8	1.436 / 1.370

Table 5. Time to apogee and maximum ascent acceleration.

Case	Motor	Time to apogee OR / RAS (s)	Peak ascent acceleration OR / RAS (m/s ²)
B01	I500T	9.195 / 8.790	78.38 / 77.64
B01	J570W	13.612 / 12.490	157.27 / 157.14
B02 *	I500T	9.098 / 8.770	78.38 / 77.64
B02 *	J570W	13.255 / 12.410	157.27 / 157.14
B03	I500T	9.258 / 9.190	78.38 / 77.64
B03	J570W	13.817 / 13.650	157.27 / 157.14
B04	I500T	17.160 / 14.080	358.30 / 354.87
B04	J570W	19.028 / 15.260	603.06 / 601.29
B05	I500T	18.373 / 17.960	358.41 / 357.32
B05	J570W	20.318 / 19.430	603.07 / 601.32

* B02 is not matched in roughness. Its numbers are included for completeness, not to rank aerodynamic solvers. Peak acceleration is especially sensitive to output sampling and should not be interpreted as independently validated loading. Native RAS summary apogees agree with all ten CSV-derived apogees within 0.005 m.

5 Drag-model evidence

MIT OR’s installed aerodynamic calculator was evaluated at RASAero’s reported Mach, altitude and zero angle of attack along ascent, using OR’s ISA atmosphere. The plots show the J570 coast branch above 30 m/s. Matching Mach explicitly avoids the export’s Mach/velocity sample offset. Residual atmosphere/Reynolds differences remain.

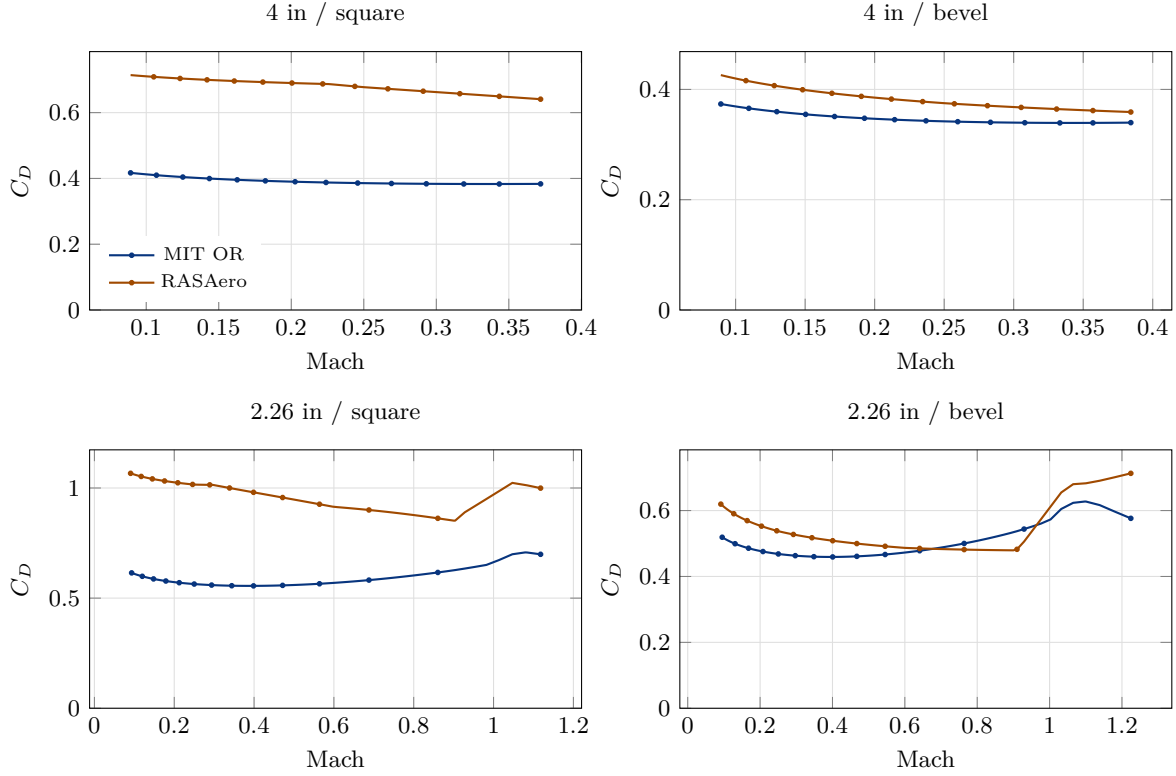


Figure 2. J570W coast-branch drag at common reported Mach and altitude, above 30 m s^{-1} . MIT OR retains its ISA atmosphere. Curves are subsampled for display only; statistics use the full histories.

Table 6. Drag coefficient at the RASAero maximum-speed row, J570W cases.

J570 case	MIT OR C_d	RAS C_d	RAS / OR
B01	0.384	0.639	1.67
B03	0.340	0.359	1.06
B04	0.588	0.920	1.56
B05	0.550	0.785	1.43

Table: coefficients at the RAS maximum-speed row, with OR evaluated at that row’s reported Mach and altitude. Both coefficients use the body frontal reference area. These are comparisons of model predictions, not measured drag.

The square-fin discrepancies persist at common flight states, so they cannot be explained solely by the trajectories reaching different speeds. Beveling changes RASAero’s predicted drag and apogee much more than MIT OR’s in these fixtures. RAS component drag decomposition was not available in the flight export. The later aerodynamic investigation uses paired finless/finned fixtures to separate the net fin contribution; its internal pressure, friction and interference terms remain unreported by RAS.

6 Attribution and numerical checks

6.1 Replace drag; retain MIT OR’s flight solver

A diagnostic listener substitutes RASAero’s trajectory-derived $C_d(\text{Mach})$ into OR’s total and axial drag coefficients. Powered and coast curves are interpolated separately. OR retains its motor, mass, atmosphere, gravity, integrator and other aerodynamic terms. This is an attribution experiment using RAS output, not an independent validation or a recommended production calibration.

Table 7. Drag-substitution diagnostic. Both differences use RASAero as denominator.

Case	Motor	Original gap vs RAS	OR + RAS drag apogee (m)	Replay residual vs RAS
B01	I500T	+7.25%	350.94	+0.050%
B01	J570W	+14.16%	780.72	+0.181%
B03	I500T	+1.00%	376.75	+0.042%
B03	J570W	+1.63%	897.78	+0.165%
B04	I500T	+32.94%	1533.69	+0.570%
B04	J570W	+38.58%	2045.57	+0.814%
B05	I500T	+2.03%	2222.26	+0.661%
B05	J570W	+8.47%	2860.33	+0.857%

This table uses RAS as the denominator in both difference columns; the flight-statistics table uses OR. All replay apogees are within 0.86% of RAS. The curve is not a complete $C_d(\text{Mach}, \text{Reynolds number}, \text{angle-of-attack})$ surface; interpolation, altitude dependence and sample timing limit interpretation of the small residual.

6.2 Checks against alternative explanations

Table 8. Numerical and input-consistency checks.

Check	Result
Shared motor curve	At common exported times: maximum thrust difference 0.00213 N (I500) and 0.00060 N (J570). Thrust RMSE below 0.00041 N.
Requested time step	Requests of 0.010 and 0.005 s do not change the actual ~ 0.0025 s step or the flight statistics. Source and installed bytecode confirm a global-step override.
Actual half-step convergence	Set the existing public timing globals from 0.0025 to 0.00125 s in the test process. Largest absolute apogee change across ten cases: 0.0163%. No application source was changed.
Windows / Mac repeat	Same MIT JAR and fixture builder: all ten baseline apogees and time histories of altitude, mass, thrust and time match exactly. Remaining compared columns differ only at floating-point roundoff ($< 8e-15$).
Output verification	All ten RAS native summary apogees agree with exported CSV peaks. Corrected CDX1 files retain the intended geometry, flow flag and launch settings.

7 Flow-setting sensitivity

The primary comparison matches fully turbulent flow. The initial RAS exports instead allow laminar flow and transition. Turning on All Turbulent Flow lowers RAS apogee by 1.7-19.5% in the clean cases; leaving that setting unmatched can make agreement look much better by cancellation.

Table 9. Sensitivity to the RASAero flow assumption.

Case	Motor	RAS default flow apogee (m)	RAS all turbulent apogee (m)	Change
B01	I500T	372.6	350.8	-5.87%
B01	J570W	870.2	779.3	-10.45%
B02 *	I500T	372.0	349.3	-6.11%
B02 *	J570W	864.6	769.9	-10.96%
B03	I500T	383.1	376.6	-1.71%
B03	J570W	921.6	896.3	-2.74%
B04	I500T	1895.1	1525.0	-19.53%
B04	J570W	2410.3	2029.1	-15.82%
B05	I500T	2341.1	2207.7	-5.70%
B05	J570W	2986.3	2836.0	-5.03%

In these default-flow sensitivity runs the beveled-fin geometry had already been corrected. "Default" refers only to the flow flag, not to an entirely uncorrected export. The manual describes RAS default transition at Reynolds number 500,000 and the checked option as immediate turbulent flow [1, pp. 55-56].

7.1 Why the roughness result is not a solver comparison

OR's NORMAL finish represents 60 micrometers. The exported RAS category, Rough Camouflage Paint, represents 30.48 micrometers [1, p. 53]. Equal category intent therefore does not give equal physical roughness. B02 remains explicitly confounded. An improvement to rough-surface aerodynamics cannot be inferred from its flight difference until physical roughness is matched.

7.2 Interpretation of the matched setting

Fully turbulent flow is chosen to match the current OR configuration, not because this study proves that every real rocket is fully turbulent from the tip. A separate measured validation should determine appropriate transition and roughness assumptions. Matching the checkbox removes one disagreement in assumptions; it does not force the two programs to use identical friction correlations.

8 Recommended MIT OR improvements

Table 10. Prioritized MIT OpenRocket improvements and acceptance evidence.

Priority / status	Change	Acceptance evidence
1 / Confirmed export issue	Map perfectFinish=false to RAS Turbulence=True. For transition-enabled OR configurations, explain that transition correlations can still differ. Location: RocketDesignDTO.	Export/reload regression fixtures verify the effective flow assumption and preserve it on save.
1 / Confirmed export issue	Map MIT TRIANGULAR leading bevel to Hexagonal Blunt Base; populate FX1 from the bevel length and preserve thickness/radius. Current mapping warns and substitutes Square. Locations: RASAeroCommonConstants, FinDTO.	Round-trip square and 0.25 in beveled fixtures; compare dimensions and native GUI geometry.
1 / Confirmed mapping limitation	Show source and destination roughness in physical units. Warn whenever no exact RAS category exists; offer explicit mapping choices. Location: surface-finish conversion / ExternalComponent.	A 60 micrometer OR finish must not be presented as numerically equivalent to 30.48 micrometers.
1 / Confirmed timing issue	Restore user/adaptive/event limits for normal flights; isolate controller timing. RK4SimulationStepper currently replaces the selected step with a global theTimeStep when no controller is active.	Tests verify requested limits, event boundaries and refinement. This fix improves timing correctness; it is not expected to close this study's drag gap.
2 / Investigated; no code changes	Prioritize tangent-ogive wave drag, fin-profile/shock treatment, component Reynolds numbers, and base drag. The fixed-Mach aerodynamic investigation below provides controlled evidence and an offline nose-term trial.	Validate proposed changes against the retained 25-case coefficient suite and independent measurements before adoption.
2 / Validation capability	Add an aerodynamic breakdown export and a reproducible benchmark suite using these fixtures: friction, pressure, base, total Cd; Mach, Reynolds, atmosphere, step size and warnings.	Track matched-case residuals and export audits in CI. Use RAS agreement as a regression signal, not as physical truth.
3 / Physical calibration research	Validate transonic/body and bevel correlations with independent flight coast-down or wind-tunnel data; examine component-local Reynolds/transition alternatives.	Hold out independent geometries/fin sections and quantify uncertainty before changing default correlations.

The code locations refer to the current MIT working tree associated with the tested application. Its source contains custom changes beyond the recorded Git HEAD; SHA-256 hashes of inspected files and installed JAR are preserved. No MIT application code was modified as part of this study.

9 Limits, evidence and reproduction

9.1 What remains unresolved

There are eight matched design/motor pairs, not eight independent physical experiments. The matrix covers only zero-wind, single-stage, axial flights and two motor curves. It does not establish CP/stability accuracy, controller behavior, staging, recovery loads or high-Mach performance. B04/B05 with J570 trigger OR’s warning that body calculations may be inaccurate at supersonic speeds.

The total drag difference is well supported. The subsequent fixed-Mach investigation identifies nose-wave and fin-profile mechanisms, while the exact RASAero split among fin pressure, friction and interference remains unresolved. Mass depletion, atmosphere/gravity, sample timing and Cd interpolation can contribute to the remaining sub-percent residual. This study does not establish that RASAero is more accurate, and it does not justify a universal multiplier on OR drag.

9.2 Execution and provenance

RASAero II runs in the Windows 11 VM (executable file/product version 1.0.2.0). Its GUI was used to edit profiles/flow settings, rerun each pair, save native CDX1 models and export flight CSVs. A per-process culture wrapper loads the installed executable unchanged. MIT OR designs and exports use the installed application’s Java APIs. The same MIT JAR was copied to the VM and executed with bundled OpenJDK 17.0.16+12-LTS for repeat verification. MIT JAR SHA-256:

d39a932bd26ba8ac9235c01332a93492f90e01f2a465e488d7fa541f31e3f4e6

Source HEAD: e8867552ba16bf53dc67ea4da55fc3c45697048b (plus local custom changes).

Table 11. Retained study files and their roles.

Study folder	Contents / usage
models/	Five ORKs with simulated data; raw exports; GUI-corrected exports. Use *-turbulent.CDX1 for the primary comparison. study-motors.eng preserves the two motor curves.
openrocket/ and rasaero/	Raw numerical histories. OR -dt005 files test the requested step; -actual-h00125 files test a true half-step; -ras-cd-replay files are the attribution diagnostic. RAS -turbulent files are primary.
evidence/	Screenshots, export warnings, model audit, common-Mach aerodynamic evaluations, runtime/source hashes, convergence logs and native-summary checks.
windows-validation/	MIT OR baseline/repeat histories and execution log from the Windows VM.
scripts/ and figures/	Fixture generator, analysis and diagnostic source, GUI automation helpers, report/plot builders and standalone scientific figures.
comparison-results.json	Machine-readable flight statistics and diagnostics; mit-or-rasaero-study.tex is the standalone report source.

For reproduction: run ControlledStudy against the installed MIT JAR, then open each export in RAS and apply the documented profile/flow corrections. Confirm the same engine entries and launch settings; rerun and export CSV at 0.01 s. Run the actual-step diagnostic, StateComparison, DragReplay and analyze_study.py. The archived UI helpers retain session-specific window coordinates/PID and need adjustment for a new desktop session.

9.3 References

[1] RASAero II User Manual: profiles p. 15, roughness p. 53, flow pp. 55-56, flight dynamics/atmosphere pp. 79-80. Official manual (PDF). Documentation was inspected from the locally retrieved manual.

[2] Installed MIT OpenRocket JAR and associated source tree; evidence/runtime.json and evidence/source-provenance.json identify the exact local artifacts. [3] Native models, raw outputs and GUI evidence in this study directory. All reported flight results are computed from those retained outputs.

10 Aerodynamic investigation and proposed corrections

10.1 Scope and controlled experiments

The follow-on investigation evaluates 25 configurations in the Windows 11 RASAero II GUI and in the installed MIT OR aerodynamic engine. It contains 7,500 common Mach/configuration points: Mach 0.01–3.00 at increments of 0.01, zero angle of attack, sea level, smooth surfaces and fully turbulent flow. These are static aerodynamic sweeps; the new coefficients have not been used to claim new flight performance. No application source, installed JAR, or original rocket model was changed.

H denotes the 4 in rocket family and S the 2.26 in family from the flight study. “sq” means square fins, “bev” means a 0.25 in leading bevel with a blunt trailing edge, and “body” means the same nose/body with fins removed. The two bevel descriptions are MIT TRIANGULAR and RASAero Hexagonal Blunt Base. Chordwise bevel distance, maximum thickness and trailing-edge thickness agree.

Table 12. Aerodynamic test matrix. All lengths below are in inches.

Family	Controlled variants	Cases
Baseline	Hsq, Hbev, Ssq, Sbev	4
Finless	Hbody, Sbody	2
Square thickness	H: 0.001, 0.0625, 0.25; S: 0.0625, 0.25	5
Fin sweep	Zero sweep on Hsq, Ssq, Hbev	3
Beveled profile	H: thickness 0.001 and 0.25; bevel length 0.50	3
Body length	Hbody: 24 and 96; Hsq: 96	3
Fin count	Hsq: three fins instead of four	1
Nose length	Hbody: 6 and 18, holding overall length at 60	2
Nozzle probe	Hbody and Sbody: nozzle diameter equal to body diameter	2

Input and extraction checks. RASAero native files are explicit variants of the already matched MIT exports. Its Aero Plots CSV concatenates 0, 2 and 4 degree sweeps; only the 2,500 zero-angle rows are selected before interpolation to the 300 OR Mach points. At zero angle, the exported CD agrees with power-off CD. In the two nozzle probes only power-on CD changes. Full-size nozzles are coefficient diagnostics, not proposed motor hardware.

The OR fixture reads the same CDX dimensions into in-memory components, calls the installed aerodynamic calculator and verifies that the component coefficients sum to the total. It uses the actual exported body length (47.9999 in for H), not a rounded replacement. Mass, motor and recovery fields are irrelevant to these coefficient-only calculations. RASAero has a 1.2384% larger body-length Reynolds number at the common nominal atmosphere; analytically matching this Reynolds number changes OR total C_D by at most 0.000775 for $M \geq 0.1$ across this suite. This cannot account for the observed discrepancies.

10.2 Separate body drag from the net fin contribution

For each geometry, define the measured finite difference

$$\Delta C_{D,\text{fins}} = C_D(\text{nose} + \text{body} + \text{fins}) - C_D(\text{nose} + \text{body}). \quad (2)$$

It includes any body/fin interference or change in exposed body surface. No unsupported identification of this quantity with pure fin pressure drag is made.

Table 13. Decomposition at Mach 0.30. Differences are RASAero minus MIT OR.

Family	Contribution	MIT OR	RASAero	Difference
H	Finless body	0.29586	0.24524	−0.05062
H	Square fin addition	0.08666	0.41220	+0.32555
H	Beveled fin addition	0.04274	0.12058	+0.07783
S	Finless body	0.37812	0.32453	−0.05359
S	Square fin addition	0.17333	0.65893	+0.48560
S	Beveled fin addition	0.07401	0.18311	+0.10910

The small total discrepancy for Hbev is a cancellation: a -0.05062 body difference plus a $+0.07783$ net-fin difference leaves only $+0.02722$. Its close subsonic flight agreement is therefore not evidence that each aerodynamic component is accurate. At Mach 0.8 the body differences are -0.11084 (H) and -0.11216 (S); at Mach 2 they reverse sign to $+0.11848$ and $+0.06843$.

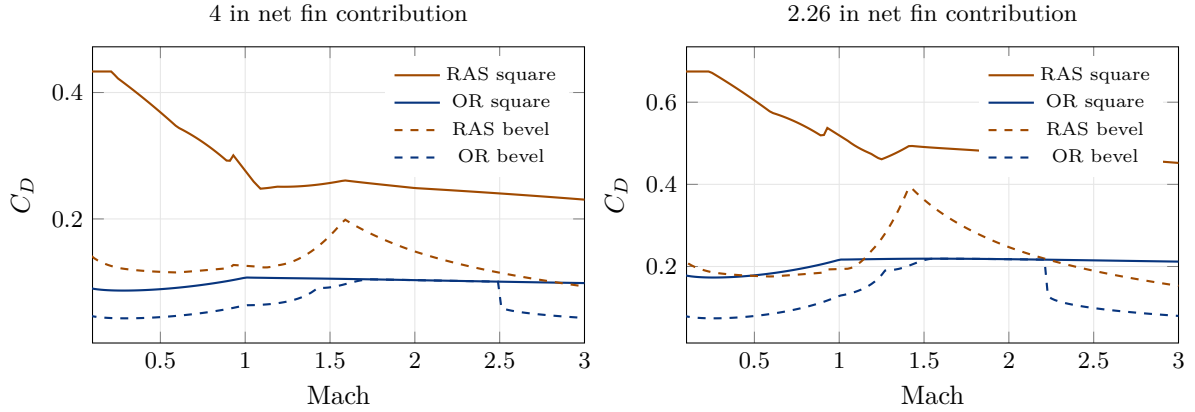


Figure 3. Paired subtraction isolates the net effect of adding fins, including any fin/body interaction. RASAero does not expose a pressure/friction split; these are not isolated fin-pressure coefficients.

Fin-count scaling agrees in both programs: replacing four H fins with three changes the RASAero net contribution by the expected factor 0.75, with a maximum coefficient residual below 1.6×10^{-7} over the full sweep. A fin-count convention error does not explain the gap.

10.3 Tangent-ogive nose pressure: the strongest corrective lead

The installed MIT engine returns only $C_{D,\text{nose}} = 0.000474$ for the H tangent ogive at Mach 2. Source inspection identifies the geometric input used in `SymmetricComponentCalc`: the slope is estimated over the last one percent of nose length,

$$s_{\text{tail}} = \frac{R - r(0.99L_n)}{\sqrt{[R - r(0.99L_n)]^2 + (0.01L_n)^2}}. \quad (3)$$

That value approaches zero for a nose tangent to its cylindrical body. The ogive branch then feeds it into a cone-like supersonic pressure formula. The H value is approximately 0.001622, whereas the radius/length equivalent-cone sine is 0.164399. This explains the near-zero pressure term and its very weak sensitivity to nose fineness. At Mach 2, shortening the H nose from 12 to 6 in raises RASAero body C_D from 0.33796 to 0.60324; OR changes only from 0.21948 to 0.22346.

Offline trial, with no fitted constants. As a diagnostic, retain OR friction and base drag, replace only the ogive pressure term, and evaluate the existing supersonic functional form using the overall cone angle:

$$s_* = \frac{R}{\sqrt{L_n^2 + R^2}}, \quad (4)$$

$$\hat{C}_{D,n} = 2.1s_*^2 + \frac{0.5s_*}{\sqrt{M^2 - 1}}, \quad M \geq 1.32, \quad (5)$$

$$\hat{C}_D = C_{D,\text{OR}} - C_{D,n,\text{OR}} + \hat{C}_{D,n}. \quad (6)$$

This is a post-processing hypothesis, not an installed patch or an experimentally validated ogive model. It does not use a RASAero coefficient in its calculation. No claim is made for its extension through Mach 1.

Table 14. Nose-only trial on six finless geometries, 169 Mach points each, 1.32–3.00. MAPE uses RASAero as denominator.

Geometry	Mean absolute C_D error		Mean absolute error [%]	
	Original	Trial	Original	Trial
H	0.11616	0.01571	35.71	5.03
S	0.06672	0.00924	20.70	2.99
H-short	0.11543	0.01596	39.73	5.69
H-long	0.11730	0.01536	30.10	4.13
H-nose6	0.37479	0.07304	63.63	12.66
H-nose18	0.06422	0.00919	23.75	3.52

The aggregate mean absolute coefficient error falls from 0.14244 to 0.02308 (83.8%). The stubby 6 in nose still has 12.66% mean error, so this simple trial is not a general replacement model. A dedicated tangent/secant-ogive wave-drag treatment, with a separately validated transonic join, is the preferred improvement.

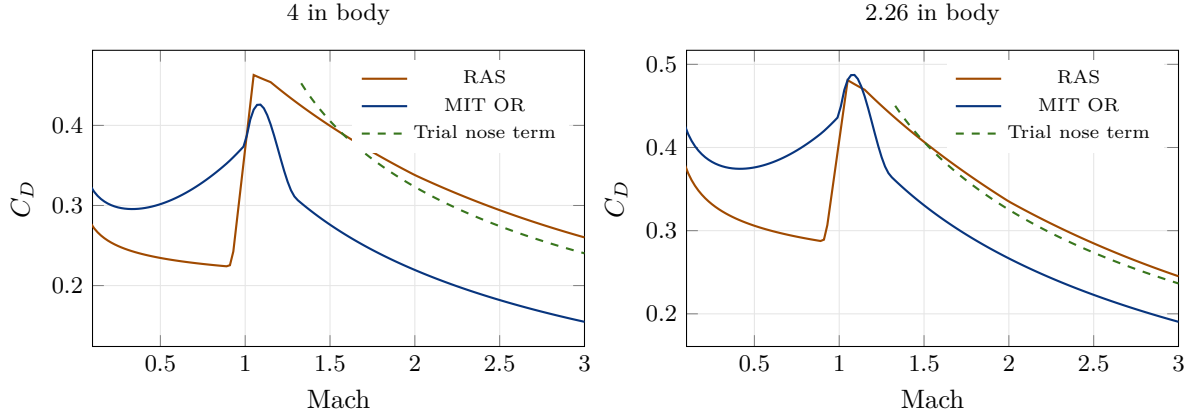


Figure 4. An offline, unfitted nose-term substitution substantially closes the supersonic finless-body gap. The trial is defined only from Mach 1.32 upward and leaves original OR friction and base drag intact.

10.4 Fin profile, thickness and sweep

The square-to-beveled change provides a strong profile diagnostic at identical planform and fin count. At Mach 0.3, the H total-coefficient reduction is 0.29163 in RASAero versus 0.04391 in OR, a factor of 6.64. At Mach 2, OR gives the same $C_D = 0.32266$ for Hsq and Hbev, while RASAero gives 0.58680 and 0.48648 respectively. Body drag cancels in these comparisons.

In **FinSetCalc**, the MIT triangular branch uses leading-edge-normal Mach, a weak attached wedge-shock calculation, and a fallback to the square-edge stagnation coefficient when no attached solution is found. That fallback produces identical square/bevel fin pressure coefficients over Mach 1.75–2.49 for H and 1.56–2.22 for S in this sweep. This loss of profile sensitivity is an implementation mechanism, not a mismatched RASAero airfoil selection. Even doubling H bevel length from 0.25 to 0.50 in leaves the OR total unchanged at Mach 2 (0.32266), while RASAero decreases from 0.48648 to 0.44915.

Proposed direction. Replace the all-or-nothing detached-shock fallback with a finite-thickness fin model that retains section geometry in attached, detached and transonic regimes. Check the coordinate projection consistently: with chordwise bevel distance b and leading-edge sweep Λ , the normal-section distance is $b \cos \Lambda$, so the normal wedge angle is $\tan^{-1}[t/(2b \cos \Lambda)]$. The present source uses the chordwise angle with normal Mach. This is a geometric consistency item to investigate, not a tested standalone correction. A normal-section wedge alone also does not account for the complete finite fin and its expansion/base regions.

Use independent sweep/profile fixtures as acceptance cases. At Mach 0.8, removing H square-fin sweep increases total C_D by 0.03781 in RASAero and 0.07748 in OR. For H beveled fins the increases are 0.01027 and 0.06198. Increasing one constant drag multiplier would not reproduce both responses.

10.5 Friction, base drag and limits of matching RASAero

Fin-local Reynolds number is supported by a length-invariance test. OR computes one turbulent skin-friction coefficient from the entire aerodynamic rocket length and applies it to the fins. Lengthening the H body from 48 to 96 in leaves the RASAero net square-fin addition unchanged at Mach 0.3 (0.41220), while OR decreases it from 0.08666 to 0.08446. Evaluating the same OR correlation on fin mean aerodynamic chord removes that artificial body-length dependence in this isolated smooth-fin model.

The candidate uses $Re_f = V\bar{c}/\nu$ in the existing fin wetted-area expression,

$$C_{D,f,\text{friction}} = C_f(Re_f, M) \left(1 + \frac{2t}{\bar{c}}\right) \frac{2NS_f}{A_{\text{ref}}}. \quad (7)$$

At Mach 0.3 it adds only 0.01515 (H) or 0.02504 (S) to OR C_D , compared with square-fin contribution gaps of 0.32555 and 0.48560. Thus it is a targeted structural improvement, not an explanation of the full discrepancy. Root boundary-layer interaction and roughness should be treated explicitly rather than inferred from a universal friction factor.

Base-drag treatment needs a separate audit. OR's body base term is $0.12 + 0.13M^2$ below Mach 1 and $0.25/M$ above it. The nearly common H/S subsonic body residual suggests a Mach-dependent body/base contribution rather than an overall geometry scale error. Changing the RASAero diagnostic nozzle diameter from zero to the full body diameter leaves power-off C_D unchanged and produces the same power-off/on difference for H and S:

Table 15. Nozzle-sensitive drag contribution compared with OR's full body-base term. The two columns are different observables, not an asserted component identity.

Mach	RAS full-nozzle $C_{D,off} - C_{D,on}$	OR body-base C_D
0.30	0.05470	0.13170
0.80	0.07919	0.20320
1.00	0.13208	0.25000
1.20	0.14780	0.20833
2.00	0.14894	0.12500
3.00	0.12360	0.08333

The export does not establish that a full nozzle removes every base-pressure effect, so the difference is not used as an absolute RASAero base coefficient. A future model should separate body base drag, fin trailing-edge drag and powered nozzle/plume effects; simply substituting this difference into OR would be unjustified. The primary flight-study nozzles remain zero.

RASAero thin-fin behavior prevents blind calibration. For H thickness 0.001 in at Mach 0.3, adding square fins changes RASAero C_D by -0.00149 , whereas the thin beveled profile adds $+0.07852$. OR gives $+0.02408$ and $+0.02372$. At ordinary square-fin thicknesses 0.0625, 0.125 and 0.25 in, RASAero's H net additions are 0.20369, 0.41220 and 0.82923: almost linear in thickness, despite unchanged planform area. The slender holdouts give the same qualitative result (0.32614, 0.65893, 1.32452). The near-zero-thickness cases are limiting diagnostics, not representative hardware. Their large profile-dependent offset, together with the unexposed RASAero component split, means an empirical fit could copy an extrapolation artifact. Do not remove finite-area skin friction merely to reproduce this limit.

10.6 Ranked aerodynamic suggestions and validation gates

Table 16. Proposed aerodynamic work. None of these changes has been applied to MIT OR.

Priority	Proposed change and reason	Evidence required before adoption
1. Nose wave drag	Replace the near-zero tail-slope input with a shape-appropriate tangent/secant-ogive pressure model. The unfitted cone-angle trial closes most of the tested supersonic body gap.	Validate several nose fineness ratios, multiple shapes, and a continuous transonic join against measured drag, retaining the six-body sweep as regression data.
2. Fin pressure / profile	Preserve bevel and thickness sensitivity through detached-shock and transonic conditions. Audit sweep projection and finite-fin pressure/expansion treatment.	Match square/bevel differences, thickness trends and independent sweep cases; check the thin-fin limit and compare with primary fin data.
3. Fin friction	Give fins a physically appropriate local Reynolds length, with an explicit root-flow treatment rather than whole-rocket-length dependence.	Preserve invariance to upstream body length in the idealized case; validate both H/S planforms and roughness separately.
4. Base drag	Audit the Mach dependence of body and fin-base pressure and introduce an explicit powered-nozzle treatment if supported by data.	Use finless models and independently known nozzle-area/plume conditions; do not infer an absolute base coefficient from the present power-on/off difference alone.
5. Component diagnostics	Expose nose pressure, body/fin friction, fin pressure, body base and fin-base terms as separate outputs with reference areas.	Require summed components to equal total drag; retain Mach, Reynolds number, angle of attack and flow settings with every output.

Do not use a universal C_D multiplier. Body residuals change sign with Mach; fin-profile effects depend on thickness and sweep; and existing subsonic bevel agreement includes cancellation. Fixing only one component can temporarily worsen total agreement. A calibration should be evaluated on geometries and Mach ranges excluded from any fitting and should retain a measured-data benchmark. The present work establishes differences between predictors, not which predictor is correct in every regime.

Validation boundary. The trial nose replacement is tested only at Mach 1.32–3 and does not validate transonic or subsonic flight improvement. In particular the heavy H flight cases never approach its validity range; their dominant target remains the fin/body subsonic balance. The 25-case sweeps exclude roughness, angle-of-attack drag, lugs, boattails, multistaging and active control. The original demonstrator is not used to infer clean component corrections because its description is more complex. No new apogee result is claimed from the offline coefficient trial.

10.7 Evidence, references and reproduction

All new fixtures, raw Windows exports, screenshots and matched coefficient tables are retained under `aerodynamic-investigation/`. The 25-entry `cases.json` records each variation. `matched-coefficients.csv` has all 7,500 matched rows; `analysis-results.json` includes extraction checks and trial-error statistics. `scripts/AeroSweep.java` calls the installed MIT engine; `scripts/analyze_aerodynamics.py` implements only analysis and the independent trial. The GUI export helper and its logs are retained with the study. These helpers are research fixtures, not modifications to OR.

The source audit used the local MIT implementations `SymmetricComponentCalc.java` (nose geometry and pressure), `FinSetCalc.java` (profile/sweep/friction), and `BarrowmanCalculator.java` (Reynolds number and base drag). The installed JAR and the seven files fingerprinted before this investigation were rechecked afterward. The preserved JAR SHA-256 is unchanged.

RASAero geometry and nozzle-input conventions are documented in the RASAero II Users Manual, pp. 14–16 and 84. The OpenRocket technical documentation, sections 3.4.2–3.4.5, gives the original skin-friction and pressure/base framework. The actual MIT source and measured program outputs take precedence when the custom triangular-fin branch differs from the historical documentation. No proprietary RASAero internal component equations are claimed from these black-box experiments.

11 Extended coefficient study and plot atlas

11.1 Completed experiments

The follow-on studies have now been run. This extension adds 67 new RASAero Windows GUI configurations to the previous 25, for 92 total. All configurations are evaluated at 300 Mach values from 0.01 to 3.00 and at 0, 2 and 4 degrees angle of attack, giving 82,800 matched coefficient points. A further 24 native OR finish sweeps verify the separate exact-roughness reconstruction. MIT OR application code and installed binaries remain unchanged.

The atlas contains **42 plot sheets with 176 panels**, plus the earlier flight and component plots. Each sheet is also supplied as a standalone SVG and PNG. The local gallery groups them by investigation topic. The retained raw exports, common-state table and bandwise error metrics support replotting without repeating the simulations.

Table 17. New configurations beyond the first 25-case study.

Study	New variations	Cases
Nose geometry	Cone, von Kármán, ellipsoid at nose lengths 6, 12, 18 in; tangent-ogive lengths 8, 24 in	11
Fin geometry	Intermediate sweep, span, taper, thickness, bevel distance and rounded profile	21
Nozzle area	Diameter fractions 0.25, 0.50, 0.75 and 1.00, with and without fins	7
Surface roughness	1.27, 6.35, 30.48, 152.4 micrometers on Hbody, Hsq and Hbev	12
Flow transition	Transitional-flow option on six H/S baseline geometries	6
Normal-force option	Rogers Modified Barrowman on four finned baseline geometries	4
Reynolds/scale	All external lengths multiplied by 0.5 and 2 on three H geometries	6

All direct aerodynamic comparisons retain the same body frontal reference area, dimensions, nominal atmosphere and angle of attack. Roughness and nozzle cases require the explicit qualifications below. The original 25 cases are re-used from preserved RASAero exports and independently re-evaluated in OR at all three angles. These are deterministic model comparisons, not 82,800 independent physical experiments.

11.2 Coefficient definitions and checks

OR's stored drag term is not wind-axis drag at nonzero angle of attack. Its simulation applies axial force C_A and normal force C_N in rocket coordinates. We therefore compare the common wind-axis quantities

$$C_D = C_A \cos \alpha + C_N \sin \alpha, \quad (8)$$

$$C_L = C_N \cos \alpha - C_A \sin \alpha. \quad (9)$$

The raw RASAero CD Power-Off column satisfies the first identity. For nonzero nozzle area, its single CL column satisfies the *power-on* identity; coast lift is reconstructed using CA Power-Off. Thus the CL comparison uses power-off lift consistently. CP is distance from the nose tip divided by body diameter; no mass/CG is used. Normal-force slope is the same $C_N(4^\circ)/(4\pi/180)$ secant in both programs. OR warns that its body normal-force treatment has limitations above Mach 1.1; the supersonic coefficient comparisons are retained to show that limitation.

11.3 What the additional data establish

The supersonic nose issue is shape-specific. At Mach 2 and nose $L/D=3$, the conical, von Kármán and ellipsoidal controls agree far more closely than the tangent ogive. This supports the earlier local-tail-slope diagnosis rather than an across-the-board error in reference area or body friction.

Table 18. Finless-body C_D at Mach 2, nose length 12 in, diameter 4 in, total length 60 in.

Nose shape	MIT OR	RASAero	RAS minus OR
Tangent ogive	0.21948	0.33796	+0.11848
Cone	0.31979	0.32754	+0.00775
von Kármán	0.30961	0.31984	+0.01023
Ellipsoid	0.38036	0.37637	-0.00399

The nose-shape curves also show that agreement at Mach 2 does not imply a correct transonic join. Several OR curves begin their rise earlier and spread it across a wider Mach interval; the RASAero rise is sharper. The atlas separates full-range and transonic behavior so the nose change is not evaluated solely on one supersonic point.

Fin discrepancies depend on the full section and planform. The thickness, sweep, span, taper and bevel-length sweeps test the earlier profile diagnosis on independent geometries. They retain large square-fin discrepancies and expose the attached/detached-shock changes in the MIT bevel branch. Rounded fins provide an additional profile control. Finless subtraction and geometric scale/length tests are plotted separately so an apparent improvement in total drag is not mistaken for agreement of each component.

A physical roughness match does not eliminate the roughness-model difference. RASAero’s available values differ from OR’s native enum values. Rather than call unlike values equivalent, the gray curves show native OR at the nearest available finish, while the blue exact-roughness curves evaluate the existing OR friction correlation at the RASAero numerical roughness. This reconstruction changes no installed code and reproduces all 24 retained native nonzero-finish sweeps to within 1.5×10^{-12} in C_D .

For smooth-baseline friction $C_{D,f,0}$ and whole-body length L , the reconstruction is

$$C_D(k) = C_D(0) + C_{D,f,0} \left[\frac{\max\{C_{f,0}, 0.032(k/L)^{0.2} f_R(M)\}}{C_{f,0}} - 1 \right], \quad (10)$$

where f_R is the installed OR roughness compressibility factor and $C_{f,0}$ its fully turbulent smooth coefficient. It preserves OR pressure/base terms. In the Hbody case at $k = 30.48$ micrometers and Mach 0.3, OR’s drag increase is 0.0453 versus 0.0305 in RASAero; at Mach 2 it is 0.0280 versus 0.0362. The residual is therefore not fixed by matching surface names or by one roughness multiplier.

The net square-fin roughness response vanishes in RASAero over this matrix. For every tested nonzero roughness and all 300 Mach values, the roughness-induced total-CD increment of Hsq equals that of Hbody within 2.1×10^{-15} . Thus the isolated square-fin contribution does not respond to roughness, while the beveled-fin contribution does. At Mach 0.3 and 30.48 micrometers, RASAero adds 0.0305 to both Hbody and Hsq but 0.0553 to Hbev. This is stronger evidence of a profile-conditional model structure than a total-drag comparison alone. It does not establish the correctness of that structure.

Flow-transition responses differ in both magnitude and sign. At Mach 0.3 the transition option changes Hsq drag by -0.23402 in RASAero versus -0.01054 in OR; for Hbev the changes are -0.04922 and -0.01054 . At Mach 2, the OR option increases Hsq and Hbev drag by 0.02668, whereas RASAero decreases them by 0.00100 and 0.00567. The OR branch uses a different supersonic compressibility factor from its fully turbulent branch; the option is not guaranteed to reduce the resulting coefficient. The net-fin transition plots remove the body response and show where this dependency enters.

The nozzle-sensitive term is area-proportional in these tests. Sweeping nozzle/body diameter ratios 0.25, 0.50, 0.75 and 1.00 gives power-off/on coefficient differences proportional to nozzle area. The fitted reference is simply the observed full-area difference, not an assumed base-pressure law. The maximum coefficient residual is below 1.2×10^{-15} over all Mach points for both the finless and square-finned families. OR does not implement the corresponding nozzle-area input in this aerodynamic fixture; its unchanged curve is shown as a missing dependency, not a matched powered-flow prediction. These data still do not identify the absolute unpowered body-base coefficient.

11.4 Normal force, CP and interpretation limits

The default RASAero Barrowman option omits subsonic body lift that OR includes. Enabling Rogers Modified Barrowman is therefore a meaningful separate normal-force comparison. It is not a substitute for the turbulent-flow option and does not change the geometric fin profile. The following table gives a common subsonic state; the atlas then tracks the behavior through Mach 3.

Table 19. Normal force and CP at Mach 0.3 and angle of attack 4 degrees. CP below is in inches from the nose tip.

Case	Quantity	MIT OR	RAS default	RAS modified
Hsq	C_N	0.6679	0.5671	0.8297
	CP [in]	42.173	43.778	44.382
Hbev	C_N	0.6679	0.5671	0.8297
	CP [in]	42.173	43.778	44.382
Ssq	C_N	0.8803	0.7340	1.1799
	CP [in]	36.681	38.541	38.944
Sbev	C_N	0.8803	0.7340	1.1799
	CP [in]	36.681	38.541	38.944

For these four cases, OR lies between the default and modified RASAero normal-force predictions at Mach 0.3. The modified option therefore does not automatically improve agreement. It also moves the RASAero CP farther aft, while OR predicts a more forward CP. These are separate force/interference-model differences, not fin-profile export errors.

Scope of inference. The measurements here are outputs of two simulation programs, not wind-tunnel or flight measurements. The studies characterize predictor disagreement and isolate dependencies; they do not establish physical truth. The new cone/Haack/ellipsoid controls support a shape-specific nose correction. The roughness, transition and fin sweeps show why a single calibration constant is inadequate. Angle-of-attack results add independent force/CP differences that a zero-angle drag fit cannot correct.

No aerodynamic correction was installed in MIT OR. The earlier nose-term trial remains an offline diagnostic, valid only over its stated Mach interval. Replacing the production transonic or fin model still requires a justified correlation and external validation. The thin-fin extrapolation remains a model-limit diagnostic rather than a physical drag target.

Numerical evidence and preservation. Every OR row passes the component-sum check. RASAero output is split by angle of attack before interpolation, and force-axis identities are checked across all matched rows. Native OR roughness curves provide an independent check of the exact-roughness reconstruction. Hashes of the installed JAR and monitored source files are retained before/after the extension. The original flight study and 25-case investigation data remain intact. One RASAero session exited during the batch; the remaining cases were run after relaunch, with an independent baseline repeat used to check continuity.

Data navigation. The `aerodynamic-investigation/extended/` directory contains 92 native RASAero input files, the raw RASAero exports, the OR engine outputs, `matched-coefficients.csv`, `error-metrics.csv`, `axis-audit.csv`, and a plot gallery in `index.html`. `figure-data.json` retains the arrays used by both the SVG/PNG figures and this standalone LaTeX atlas. Plot coordinates are adaptively simplified with interpolation error below the greater of 10^{-6} or 10^{-3} of each curve range; global extrema and a Mach spacing no larger than 0.05 are retained. All numerical metrics use the full Mach grid.

The native GUI inputs follow the RASAero II Users Manual, especially nose/profile geometry, pp. 10–16; roughness, pp. 52–53; and modified-Barrowman/flow options, pp. 54–56. The actual installed MIT source supplies the OR formulas. No black-box RASAero component coefficient is mislabeled as a published internal equation.

11.5 Coefficient plot atlas

The following sheets are grouped by mechanism. Blue dashed and orange solid normally denote MIT OR and RASAero; each sheet states its legend explicitly. Roughness reconstructions, nozzle comparisons and multi-angle plots have separately labeled lines.

Sheet	Topic
1	Baseline zero-angle drag (Fig. 5)
2	Transonic drag detail (Fig. 6)
3	Drag residuals by Mach (Fig. 7)
4	Nose shapes at $L/D = 1.5$ (Fig. 8)
5	Nose shapes at $L/D = 3$ (Fig. 9)
6	Nose shapes at $L/D = 4.5$ (Fig. 10)
7	Tangent-ogive fineness sweep (Fig. 11)
8	Sensitivity to shortening the nose (Fig. 12)
9	Nose-shape transonic detail at $L/D = 3$ (Fig. 13)
10	Body length and fin count controls (Fig. 14)
11	Square fin thickness sweep (Fig. 15)
12	Beveled fin thickness sweep (Fig. 16)
13	Leading bevel length sweep (Fig. 17)
14	Square fin sweep (Fig. 18)
15	Beveled fin sweep (Fig. 19)
16	Fin span and aspect ratio (Fig. 20)
17	Fin taper at fixed sweep (Fig. 21)
18	Net drag added by fins (Fig. 22)
19	Thin-profile limits and rounded control (Fig. 23)
20	Geometric scale and Reynolds number (Fig. 24)
21	Fin drag versus upstream body length (Fig. 25)
22	Physical roughness: Hbody (Fig. 26)
23	Physical roughness: Hsq (Fig. 27)
24	Physical roughness: Hbev (Fig. 28)
25	Roughness-induced drag at fixed Mach (Fig. 29)
26	Isolated fin response to physical roughness (Fig. 30)
27	Laminar-to-turbulent option: finned rockets (Fig. 31)
28	Sensitivity to flow-transition treatment (Fig. 32)
29	Isolated fin response to flow transition (Fig. 33)
30	Nozzle-area sensitivity: Hbody (Fig. 34)
31	Nozzle-area sensitivity: Hsq (Fig. 35)
32	Test of nozzle-area scaling (Fig. 36)
33	Wind-axis drag at 0, 2 and 4 degrees (Fig. 37)
34	Axial force at 0, 2 and 4 degrees (Fig. 38)
35	Normal force at 0, 2 and 4 degrees (Fig. 39)
36	Wind-axis lift at 0, 2 and 4 degrees (Fig. 40)
37	Center of pressure at 0 degrees (Fig. 41)
38	Center of pressure at 4 degrees (Fig. 42)
39	Normal-force slope over 0 to 4 degrees (Fig. 43)
40	Rogers Modified Barrowman: CN at 4 degrees (Fig. 44)
41	Rogers Modified Barrowman: CP at 4 degrees (Fig. 45)
42	Rogers Modified Barrowman: CD at 4 degrees (Fig. 46)

Baseline zero-angle drag

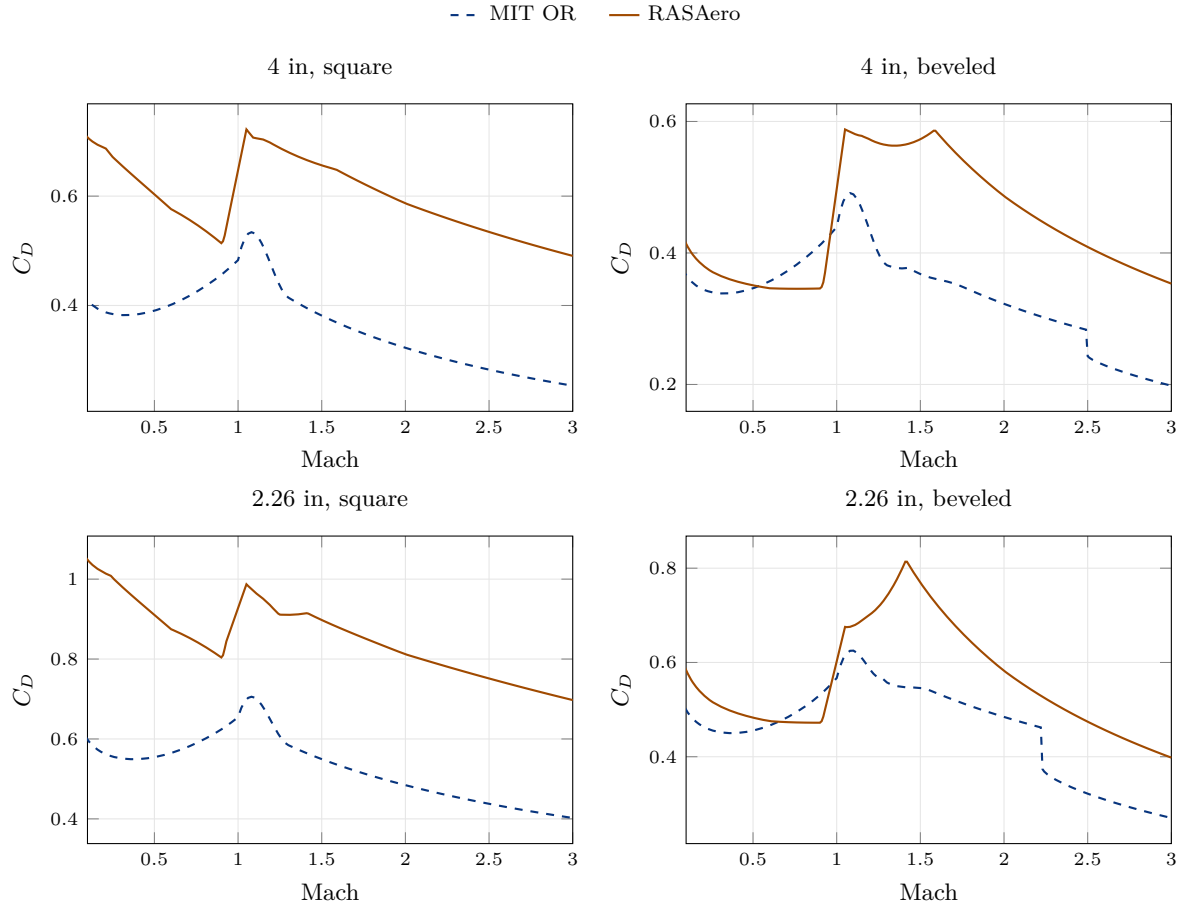


Figure 5. All surfaces have zero roughness; all-turbulent flow is selected. Direct installed-engine curves, without correction.

Export files: 01-baseline-cd.svg and 01-baseline-cd.png.

Directory: aerodynamic-investigation/extended/figures/.

Transonic drag detail

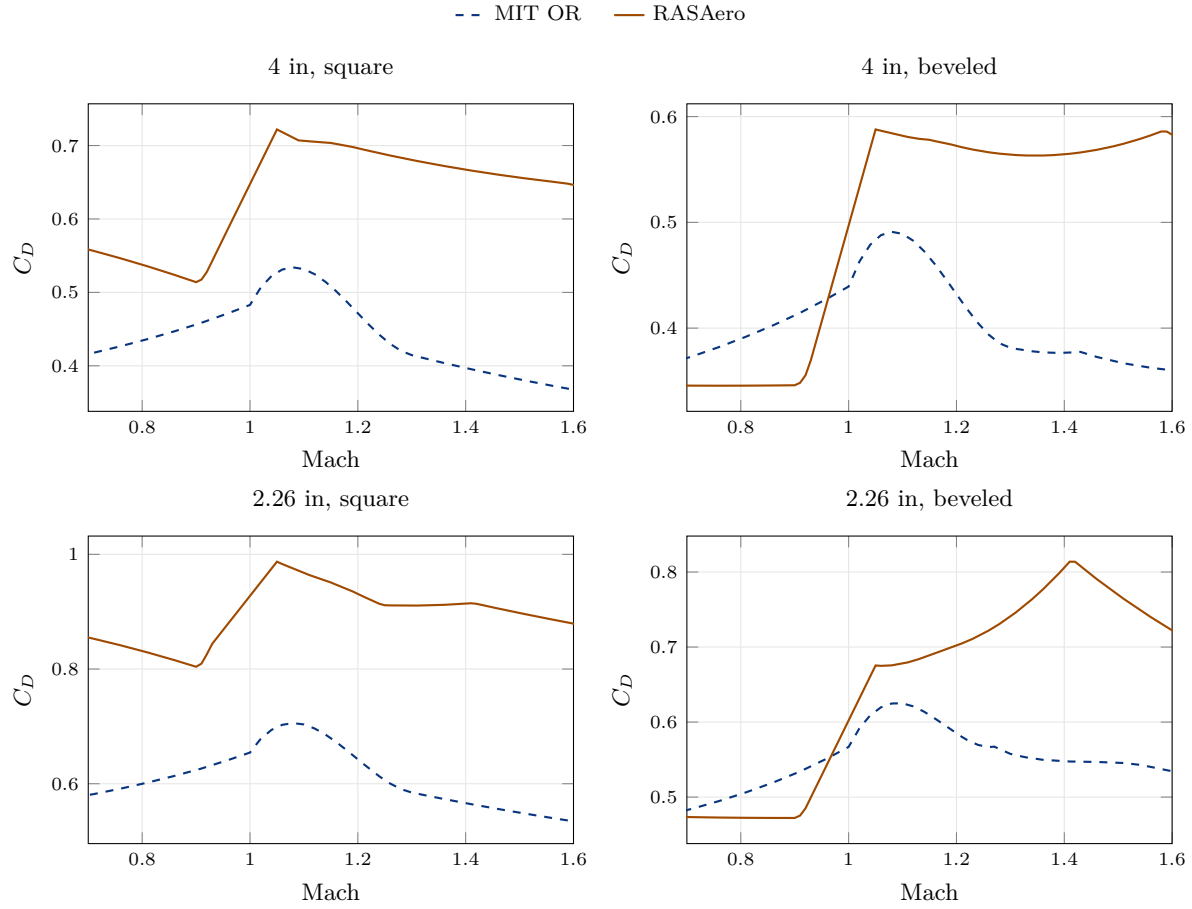


Figure 6. The denser transonic comparison tests the location, width and magnitude of the drag rise, rather than matching one peak.

Export files: 02-transonic-cd.svg and 02-transonic-cd.png.

Directory: aerodynamic-investigation/extended/figures/.

Drag residuals by Mach

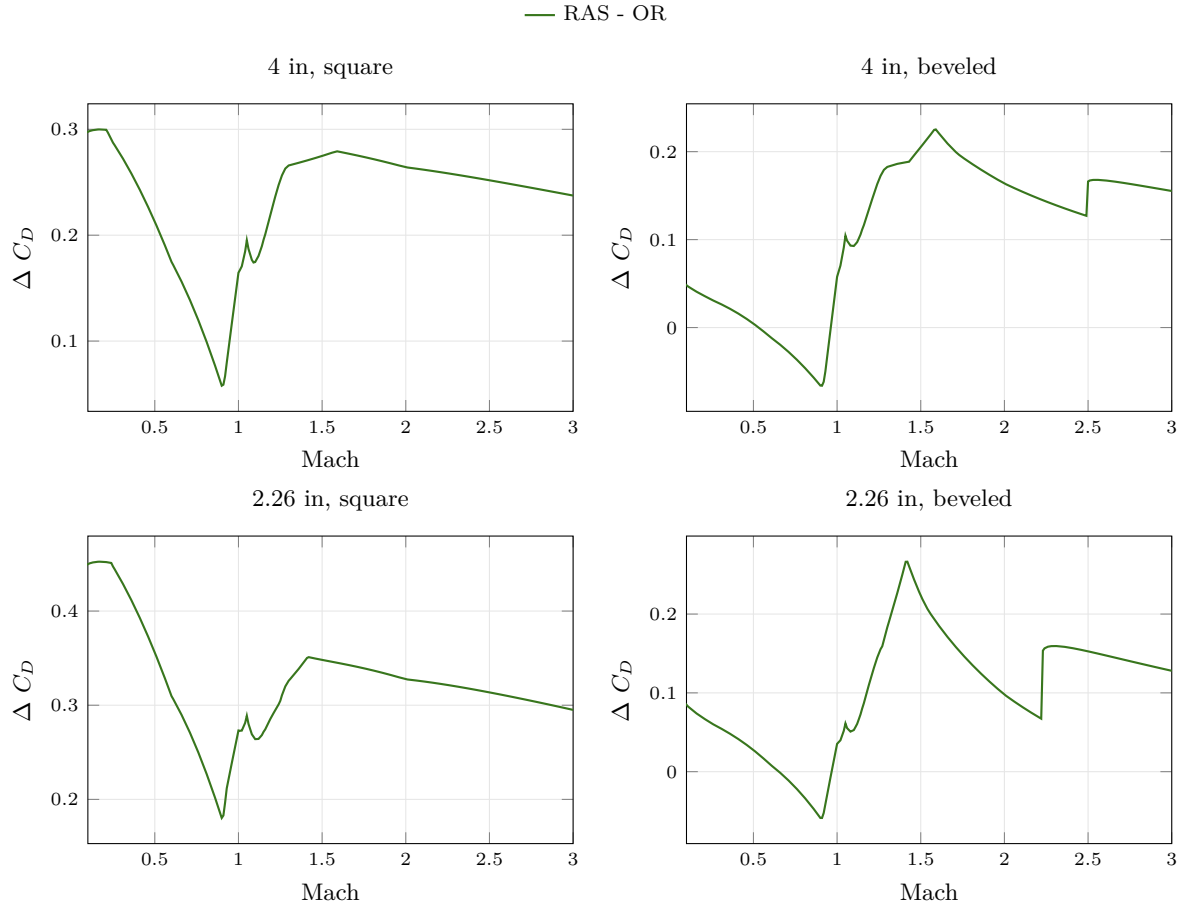


Figure 7. Positive residual means RASAero predicts more drag. The sign and size vary with geometry and Mach.

Export files: 03-baseline-residual.svg and 03-baseline-residual.png.

Directory: aerodynamic-investigation/extended/figures/.

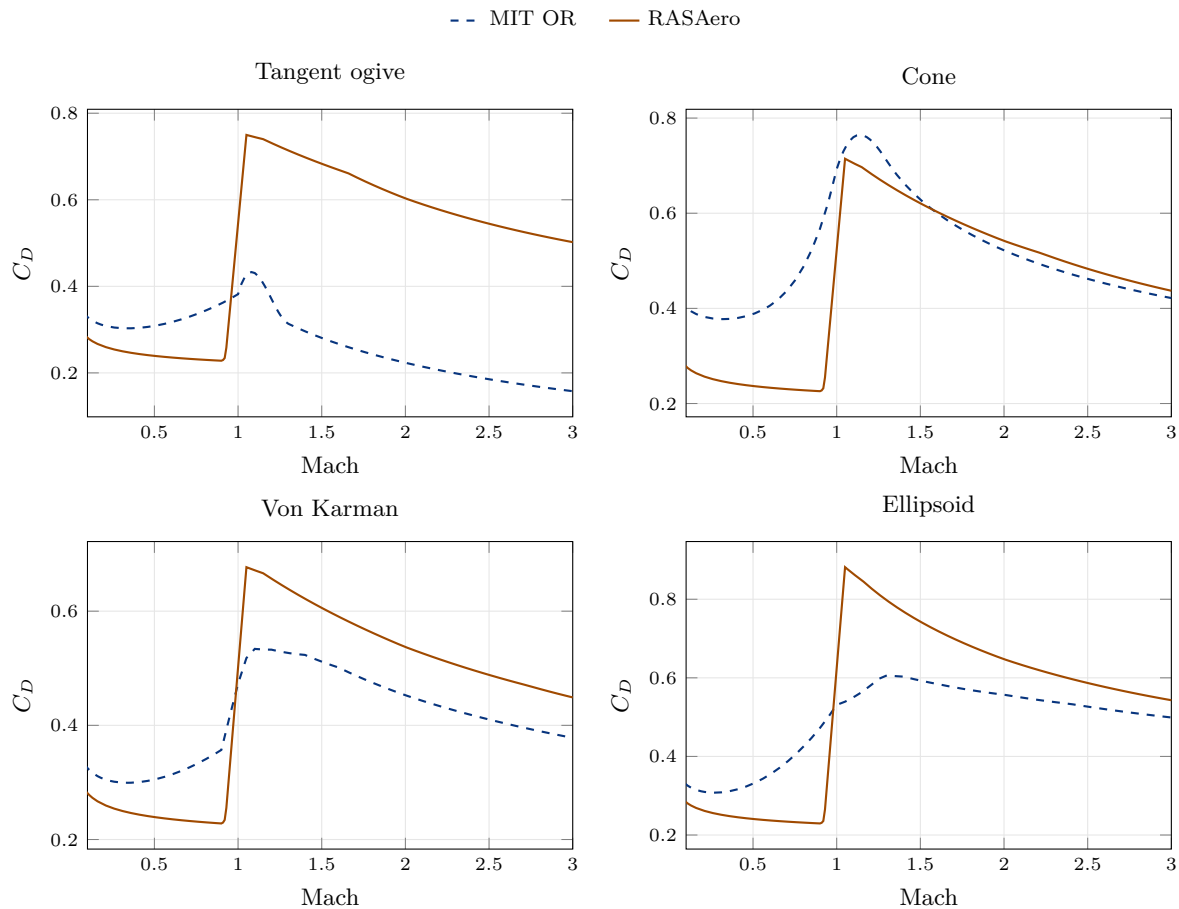
Nose shapes at $L/D = 1.5$ 

Figure 8. Finless 4 in body, overall length 60 in, nose length 6 in. Body length changes to hold total length fixed.

Export files: 04-noses-L6.svg and 04-noses-L6.png.

Directory: aerodynamic-investigation/extended/figures/.

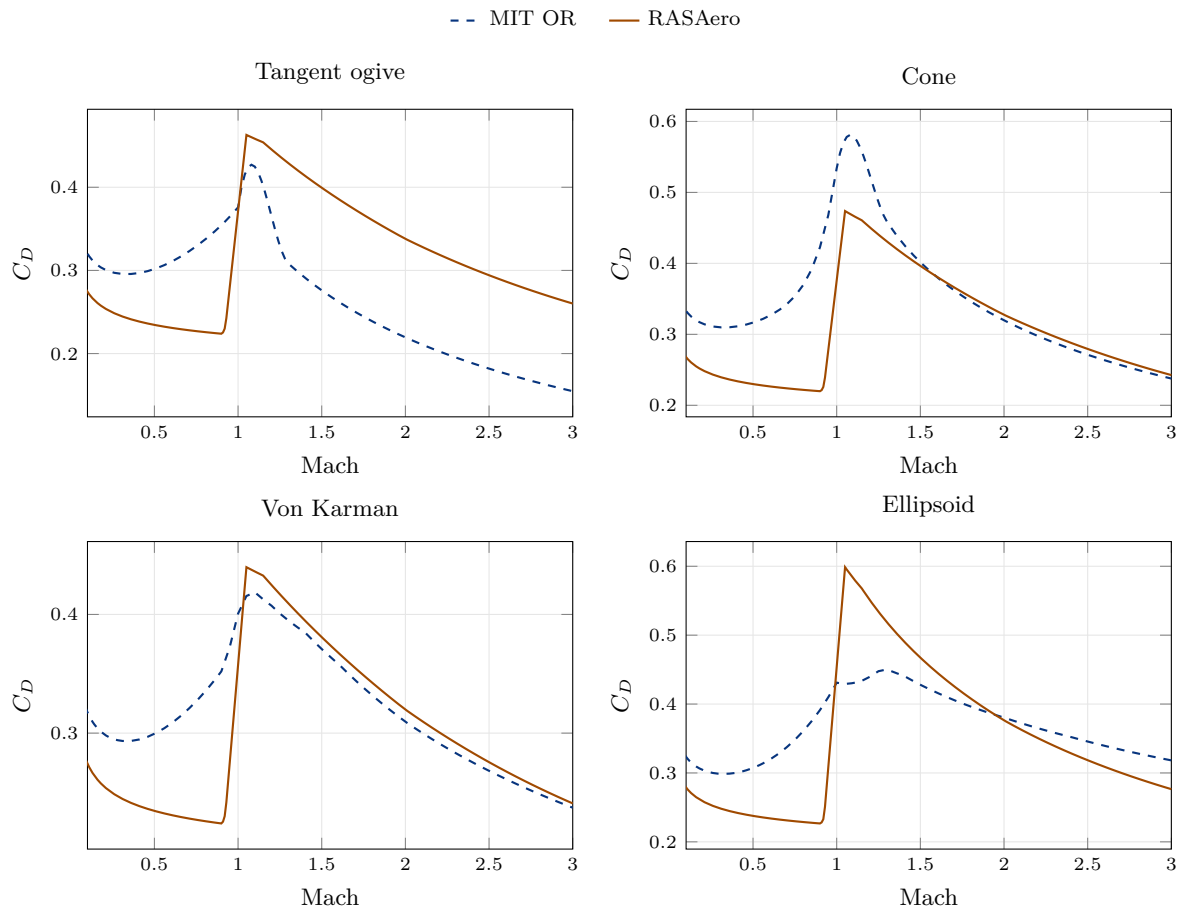
Nose shapes at $L/D = 3$ 

Figure 9. Finless 4 in body, overall length 60 in, nose length 12 in. Body length changes to hold total length fixed.

Export files: 04-noses-L12.svg and 04-noses-L12.png.

Directory: aerodynamic-investigation/extended/figures/.

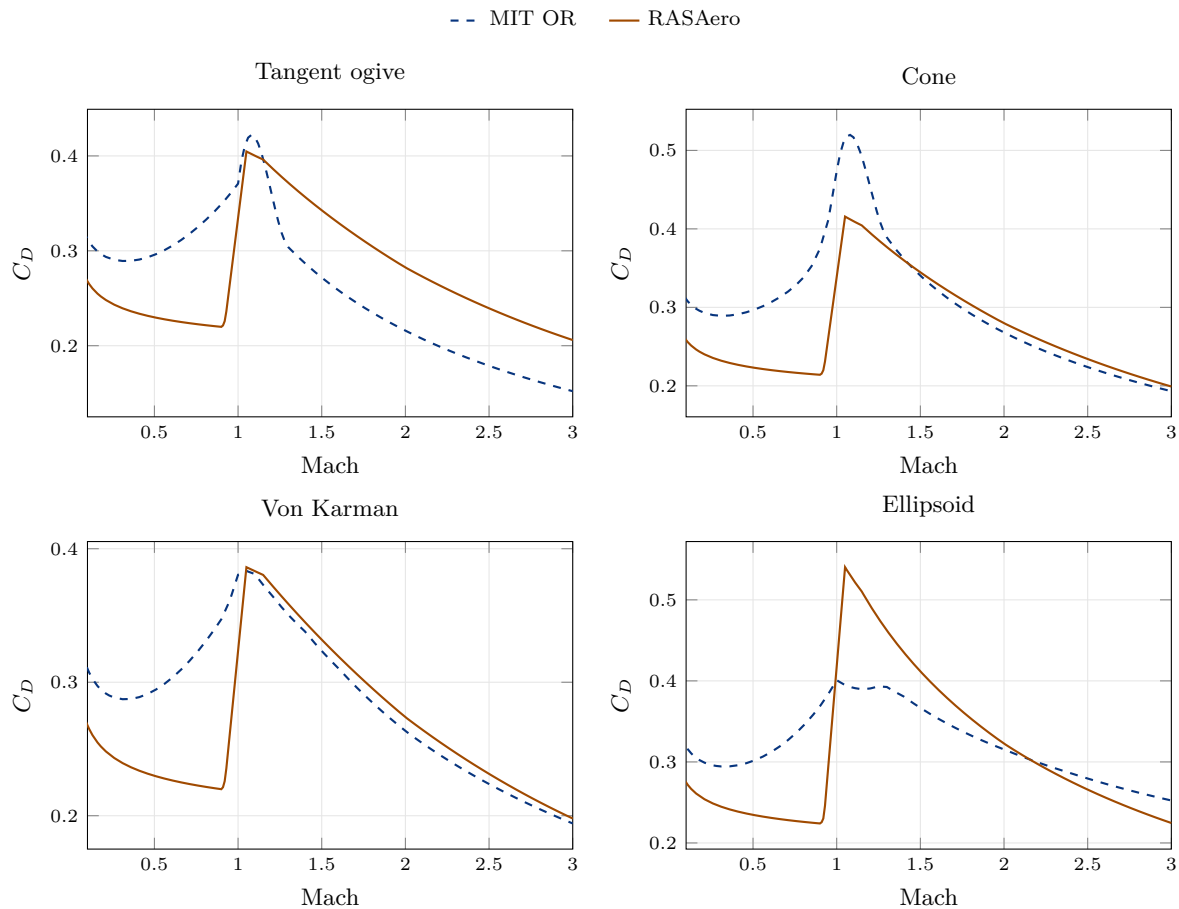
Nose shapes at $L/D = 4.5$ 

Figure 10. Finless 4 in body, overall length 60 in, nose length 18 in. Body length changes to hold total length fixed.

Export files: 04-noses-L18.svg and 04-noses-L18.png.

Directory: aerodynamic-investigation/extended/figures/.

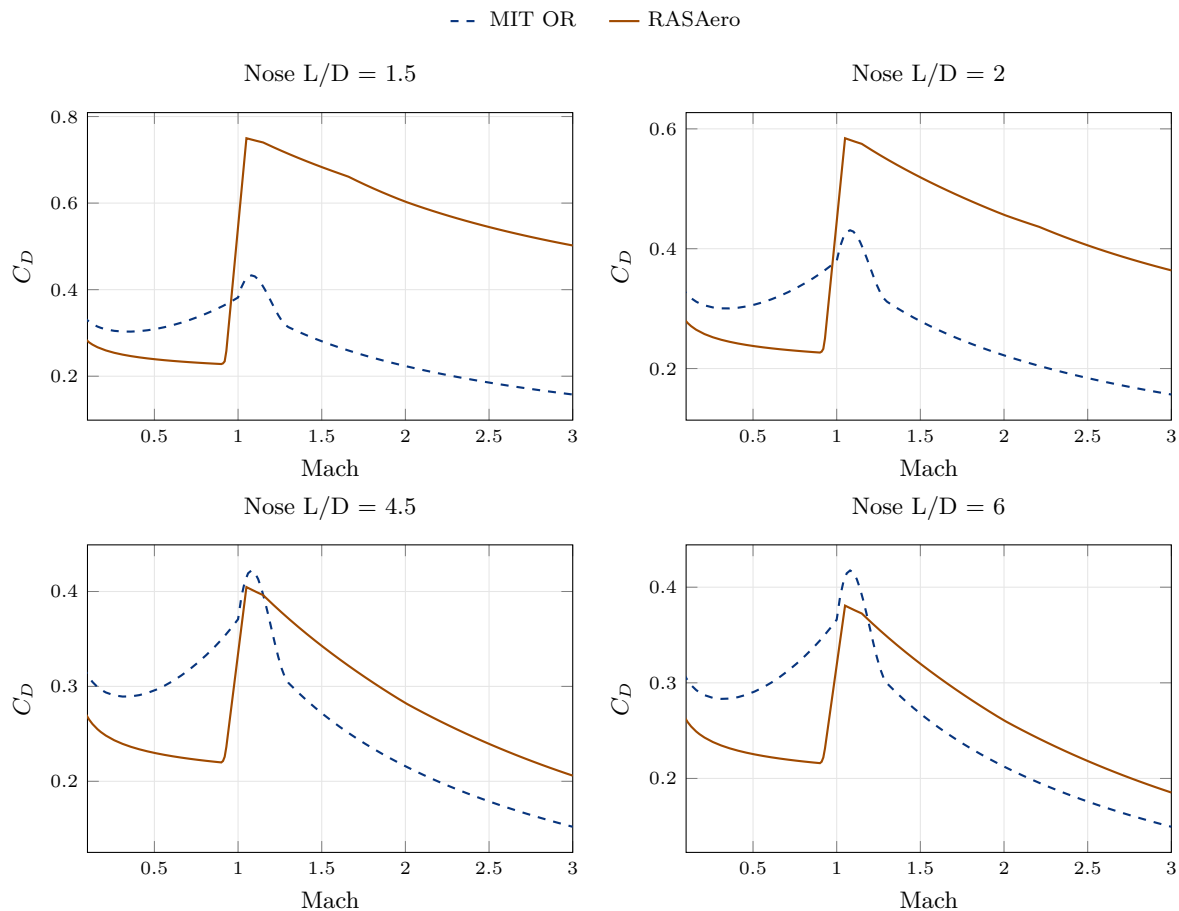
Tangent-ogive fineness sweep

Figure 11. Nose length changes at fixed overall length. L/D is nose length divided by body diameter.

Export files: 05-ogive-fineness.svg and 05-ogive-fineness.png.

Directory: aerodynamic-investigation/extended/figures/.

Sensitivity to shortening the nose

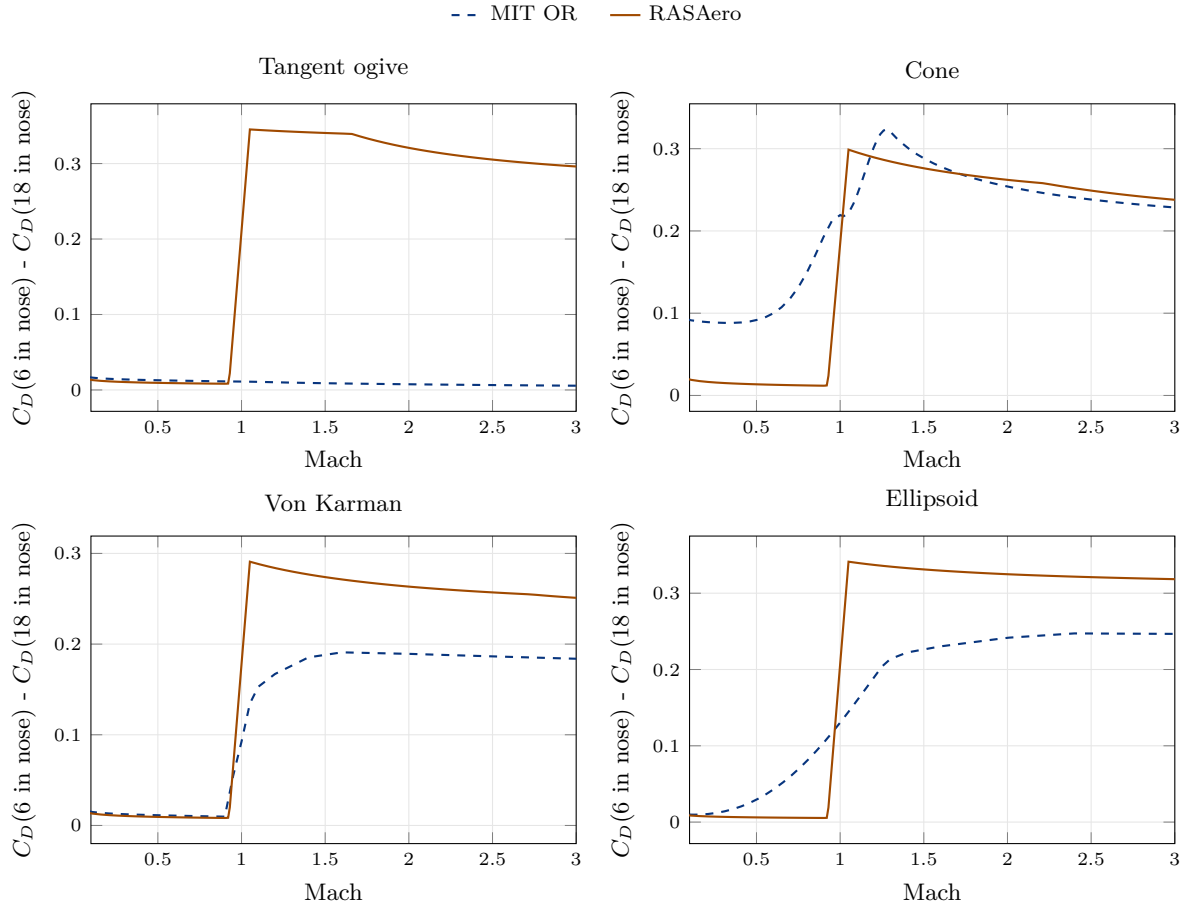


Figure 12. The finite difference contains both the nose-shape change and the associated body wetted-area change. Both solvers receive the same dimensions.

Export files: 06-nose-shape-sensitivity.svg and 06-nose-shape-sensitivity.png.

Directory: aerodynamic-investigation/extended/figures/.

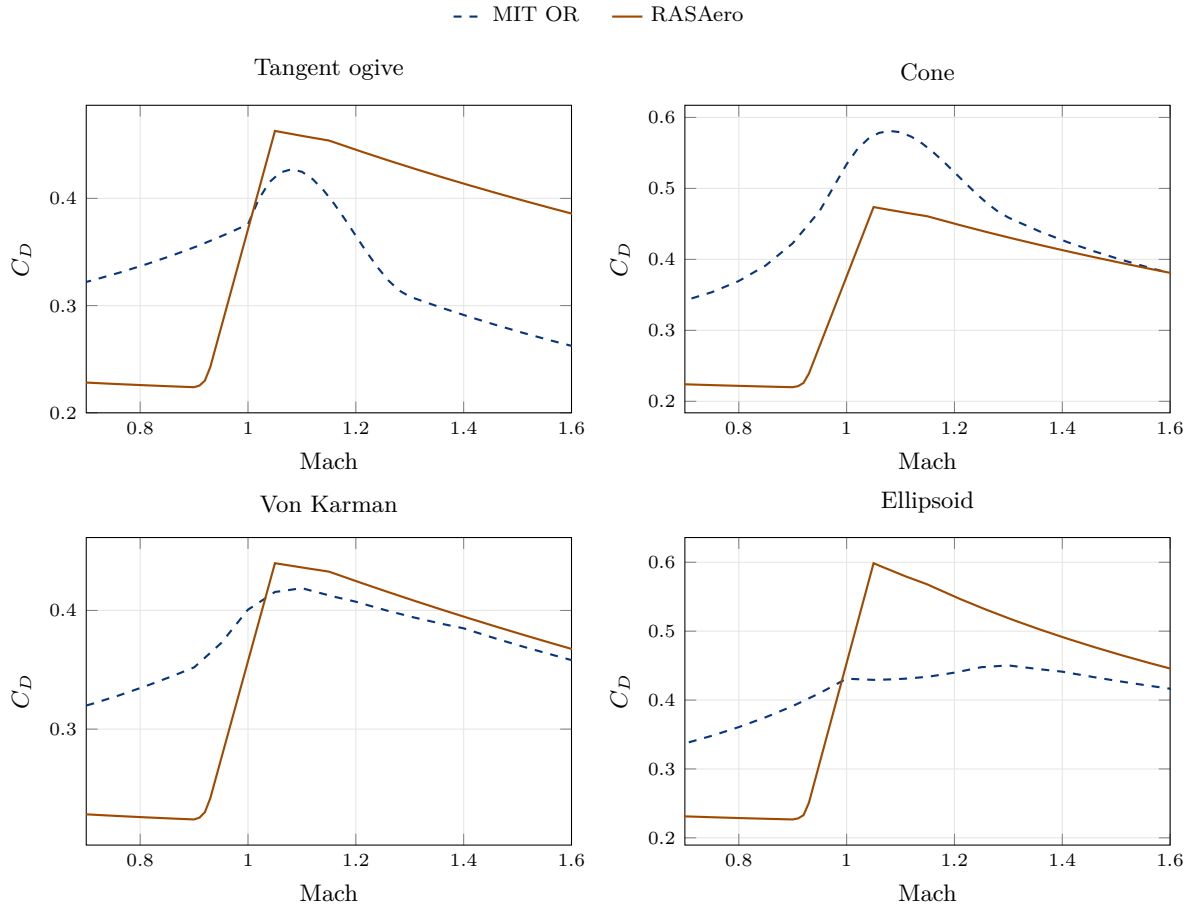
Nose-shape transonic detail at $L/D = 3$ 

Figure 13. Geometry and zero roughness are matched. Different onset, peak Mach and width of the transonic rise remain even in shapes that agree closely at Mach 2.

Export files: 29-nose-transonic-detail.svg and 29-nose-transonic-detail.png.

Directory: aerodynamic-investigation/extended/figures/.

Body length and fin count controls

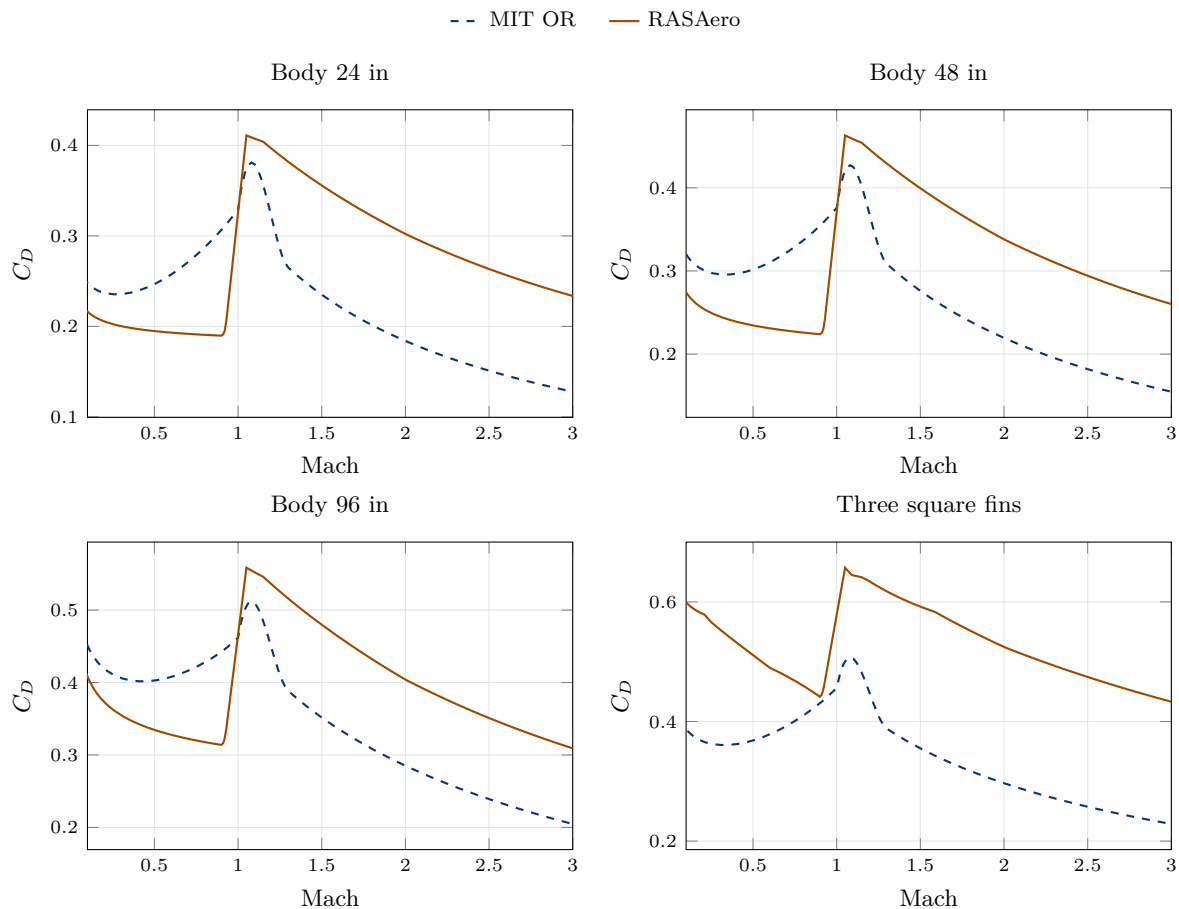


Figure 14. The first three panels vary only finless-body length. The fourth checks the 3-fin count against the original 4-fin family.

Export files: 07-body-length.svg and 07-body-length.png.

Directory: aerodynamic-investigation/extended/figures/.

Square fin thickness sweep

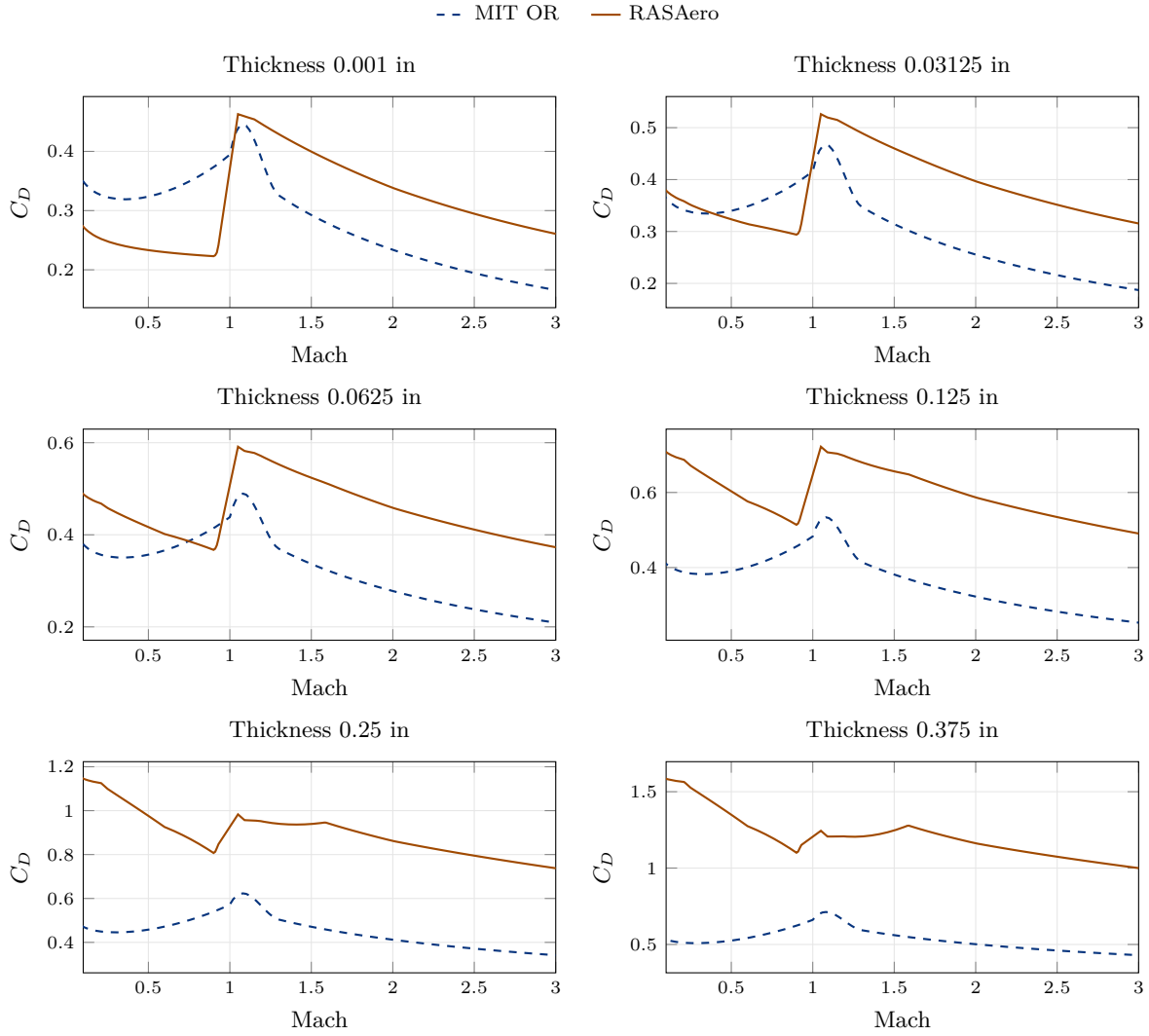


Figure 15. Fixed planform and sweep. The 0.001 in case is an asymptotic diagnostic, not a representative hardware design.

Export files: 08-Hsq-thickness.svg and 08-Hsq-thickness.png.

Directory: aerodynamic-investigation/extended/figures/.

Beveled fin thickness sweep

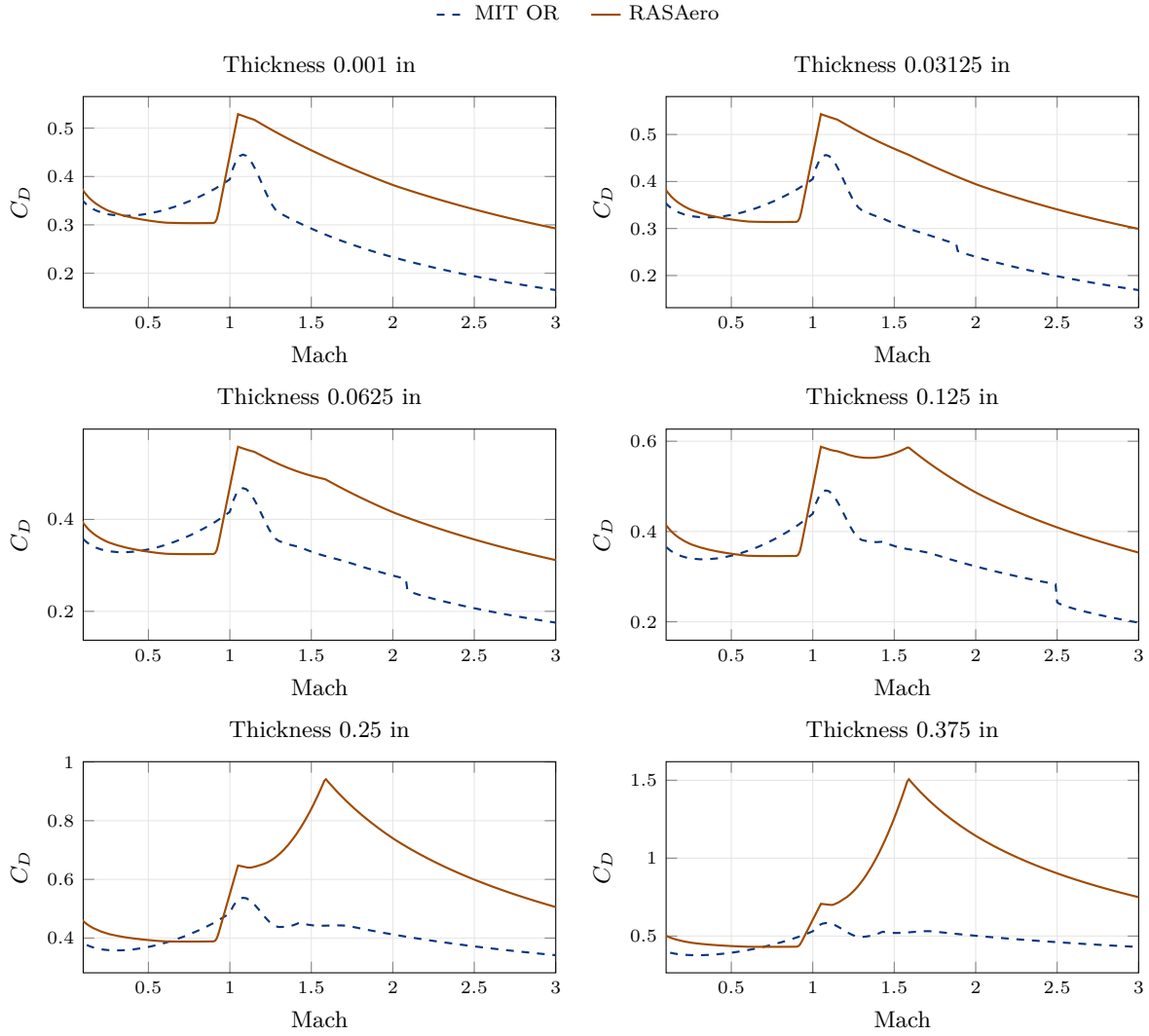


Figure 16. Fixed planform and sweep. The 0.001 in case is an asymptotic diagnostic, not a representative hardware design.

Export files: 08-Hbev-thickness.svg and 08-Hbev-thickness.png.

Directory: aerodynamic-investigation/extended/figures/.

Leading bevel length sweep

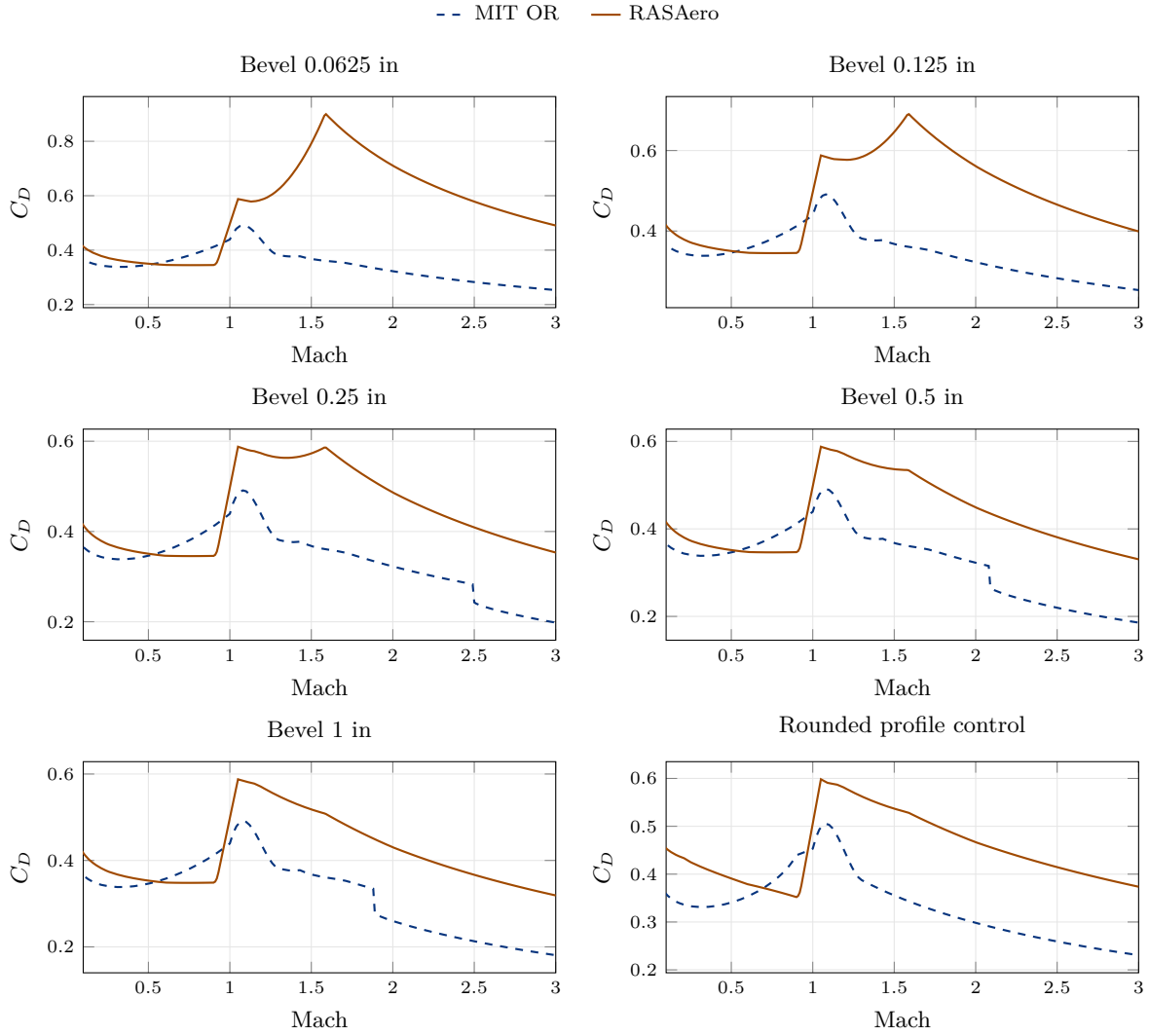


Figure 17. FX1 is measured parallel to the body. Maximum thickness is 0.125 in and the trailing edge remains blunt.

Export files: 09-bevel-length.svg and 09-bevel-length.png.

Directory: aerodynamic-investigation/extended/figures/.

Square fin sweep

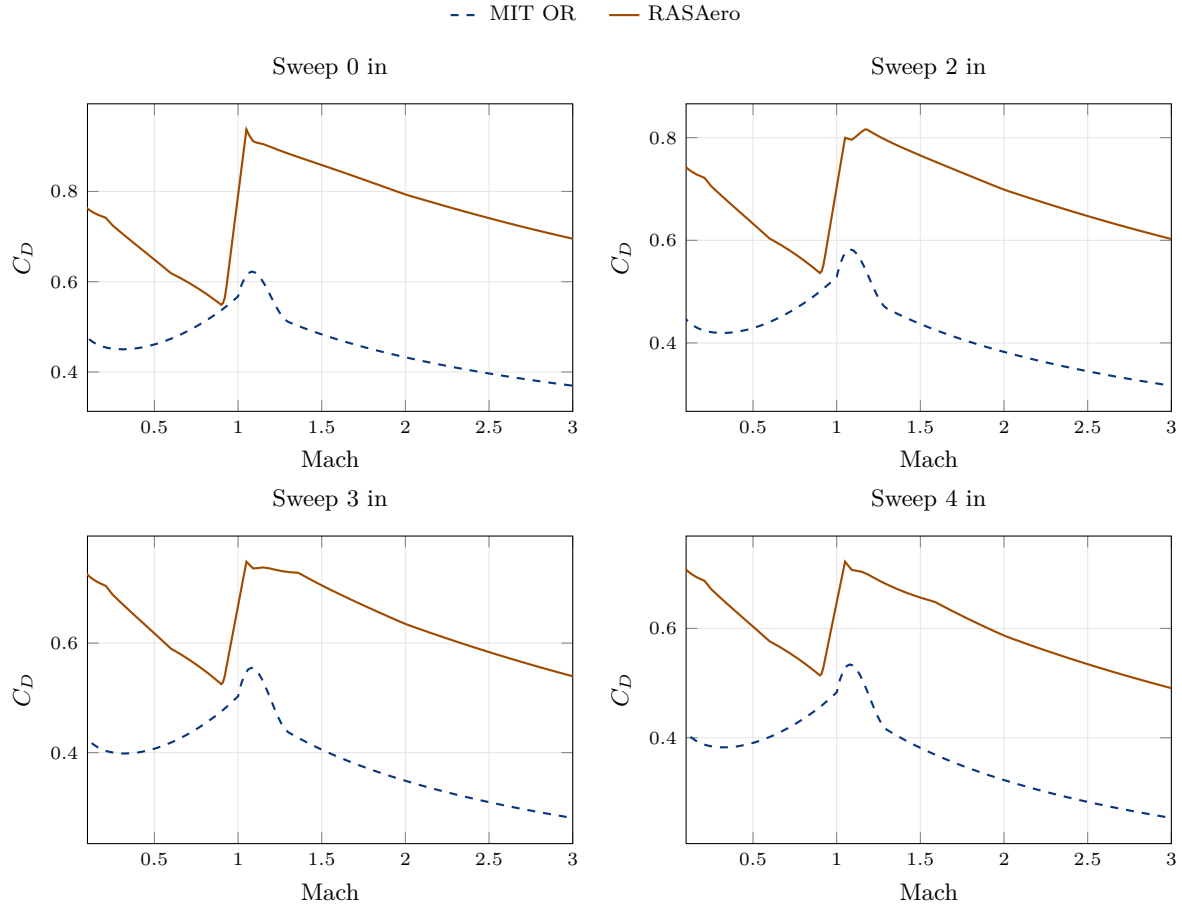


Figure 18. Root/tip chords, span and thickness are fixed. Leading-edge sweep distance varies without fin overhang past the body.

Export files: 10-Hsq-sweep.svg and 10-Hsq-sweep.png.

Directory: aerodynamic-investigation/extended/figures/.

Beveled fin sweep

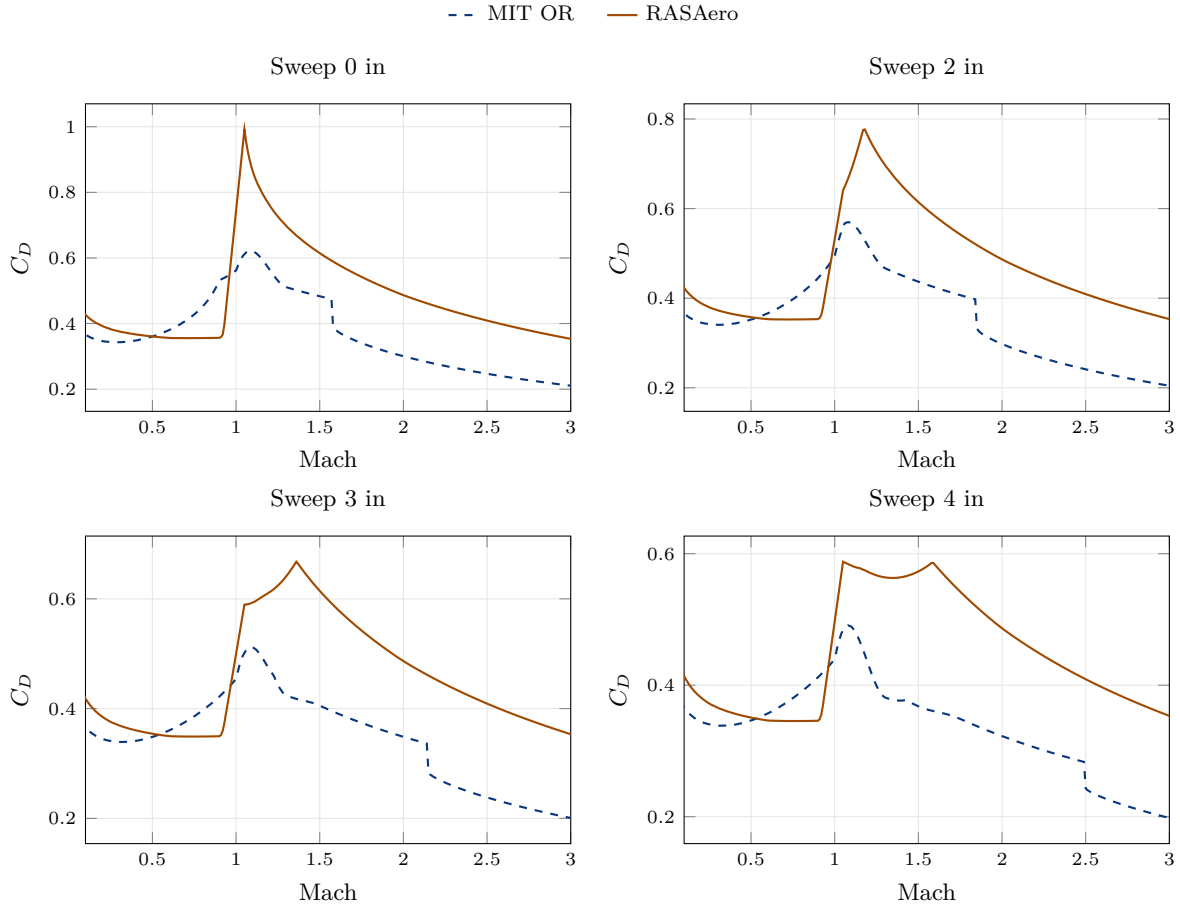


Figure 19. Root/tip chords, span and thickness are fixed. Leading-edge sweep distance varies without fin overhang past the body.

Export files: 10-Hbev-sweep.svg and 10-Hbev-sweep.png.

Directory: aerodynamic-investigation/extended/figures/.

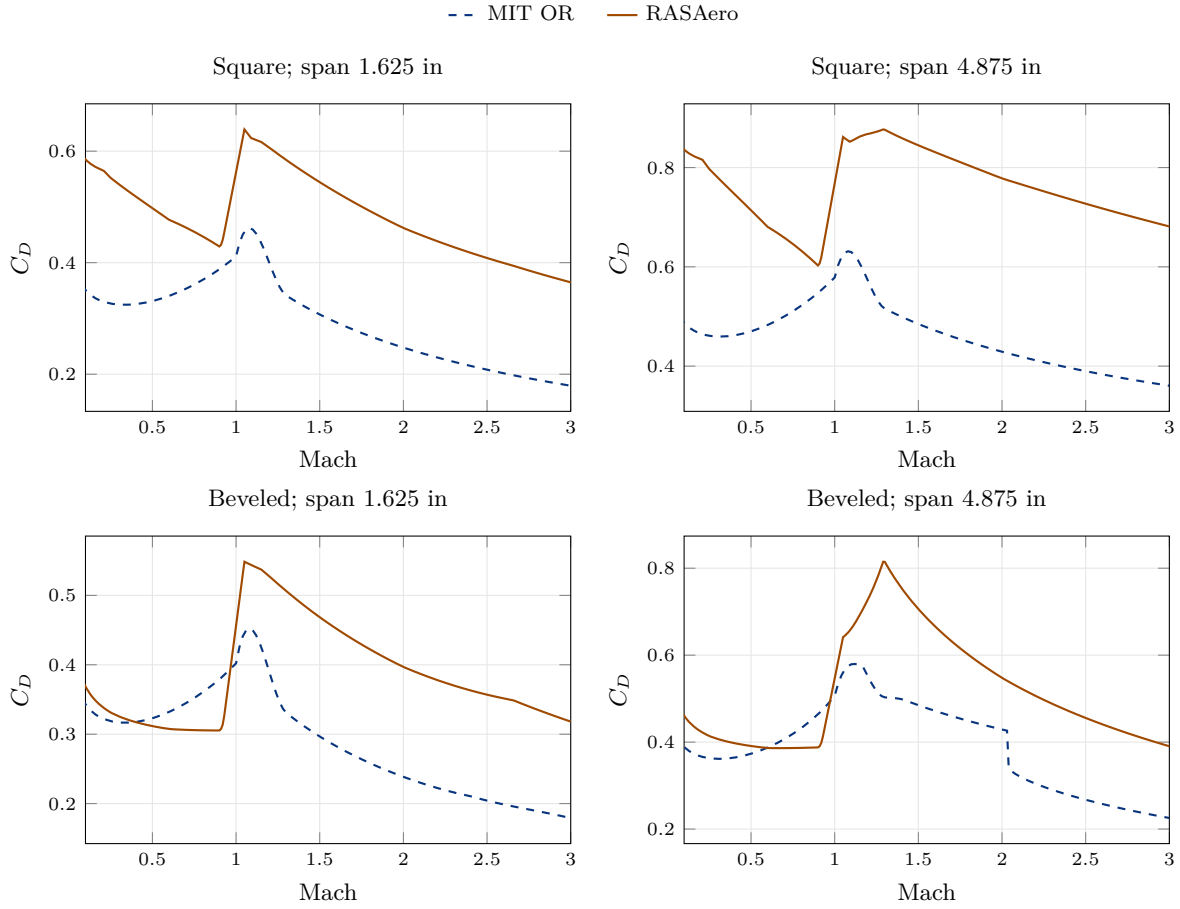
Fin span and aspect ratio

Figure 20. Root/tip chords, sweep and thickness are fixed. Each solver uses the same exposed fin span.

Export files: 11-fin-span.svg and 11-fin-span.png.

Directory: aerodynamic-investigation/extended/figures/.

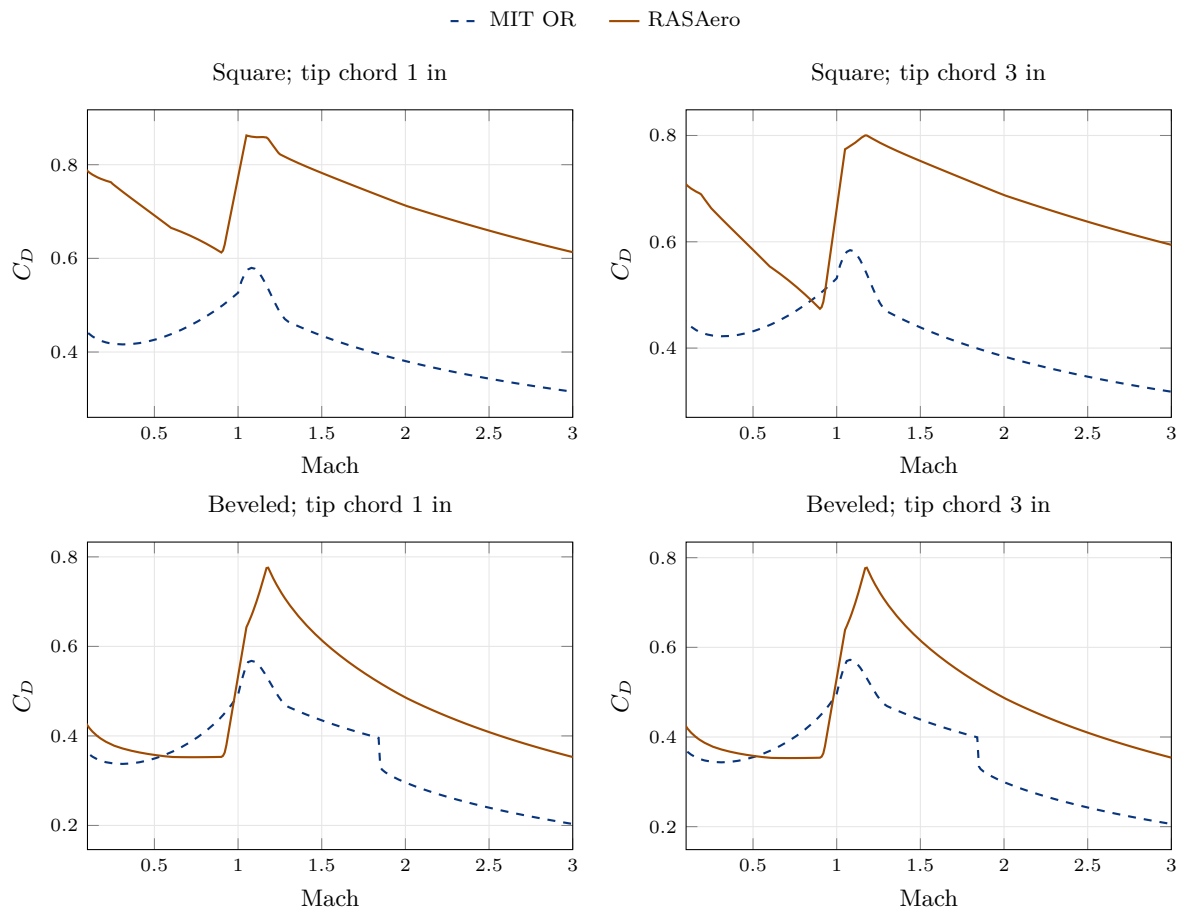
Fin taper at fixed sweep

Figure 21. Sweep is 2 in for all four cases; compare with the corresponding 2 in-sweep baseline, whose tip chord is 2 in.

Export files: 12-tip-chord.svg and 12-tip-chord.png.

Directory: aerodynamic-investigation/extended/figures/.

Net drag added by fins

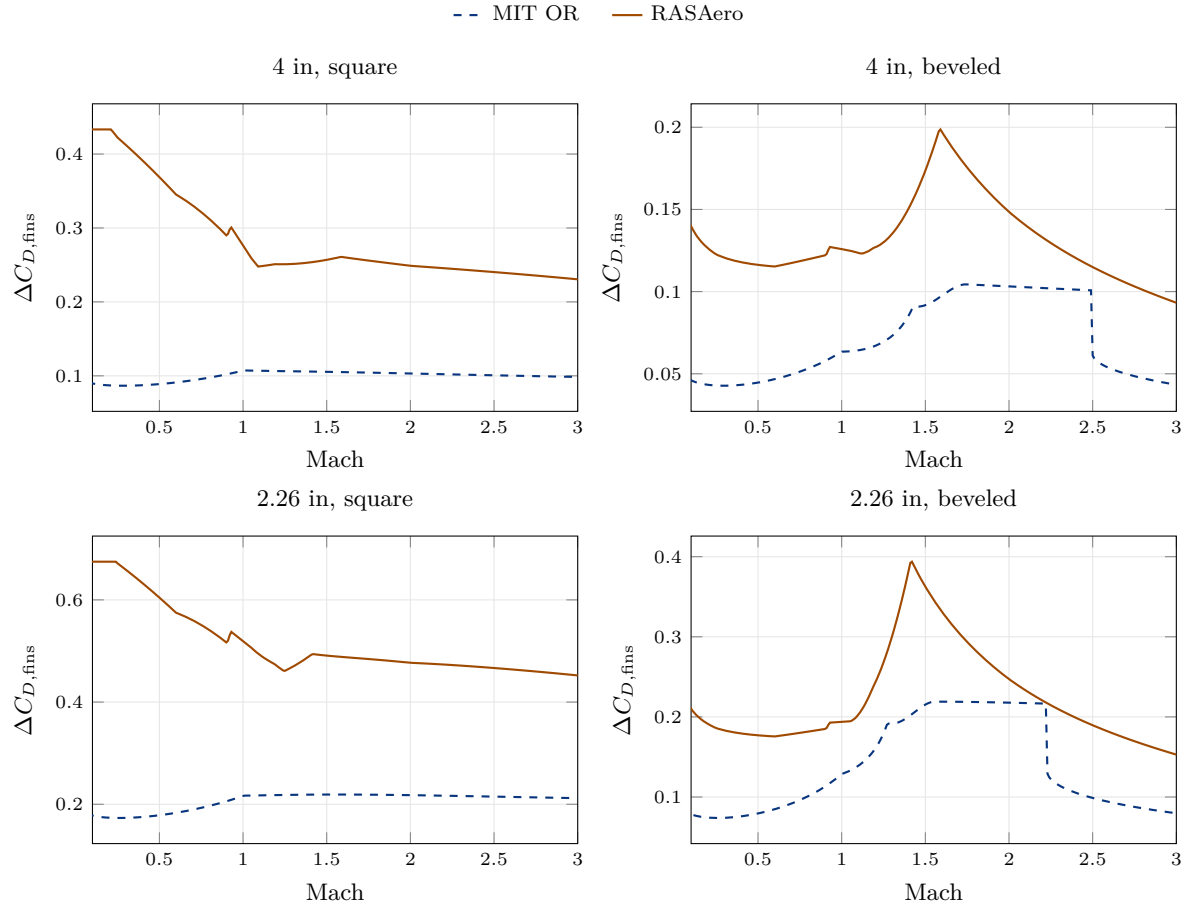


Figure 22. Finned-minus-finless subtraction includes fin/body interference and changes in exposed body surface. It is not an isolated RASAero fin-pressure coefficient.

Export files: 13-fin-increments.svg and 13-fin-increments.png.

Directory: aerodynamic-investigation/extended/figures/.

Thin-profile limits and rounded control

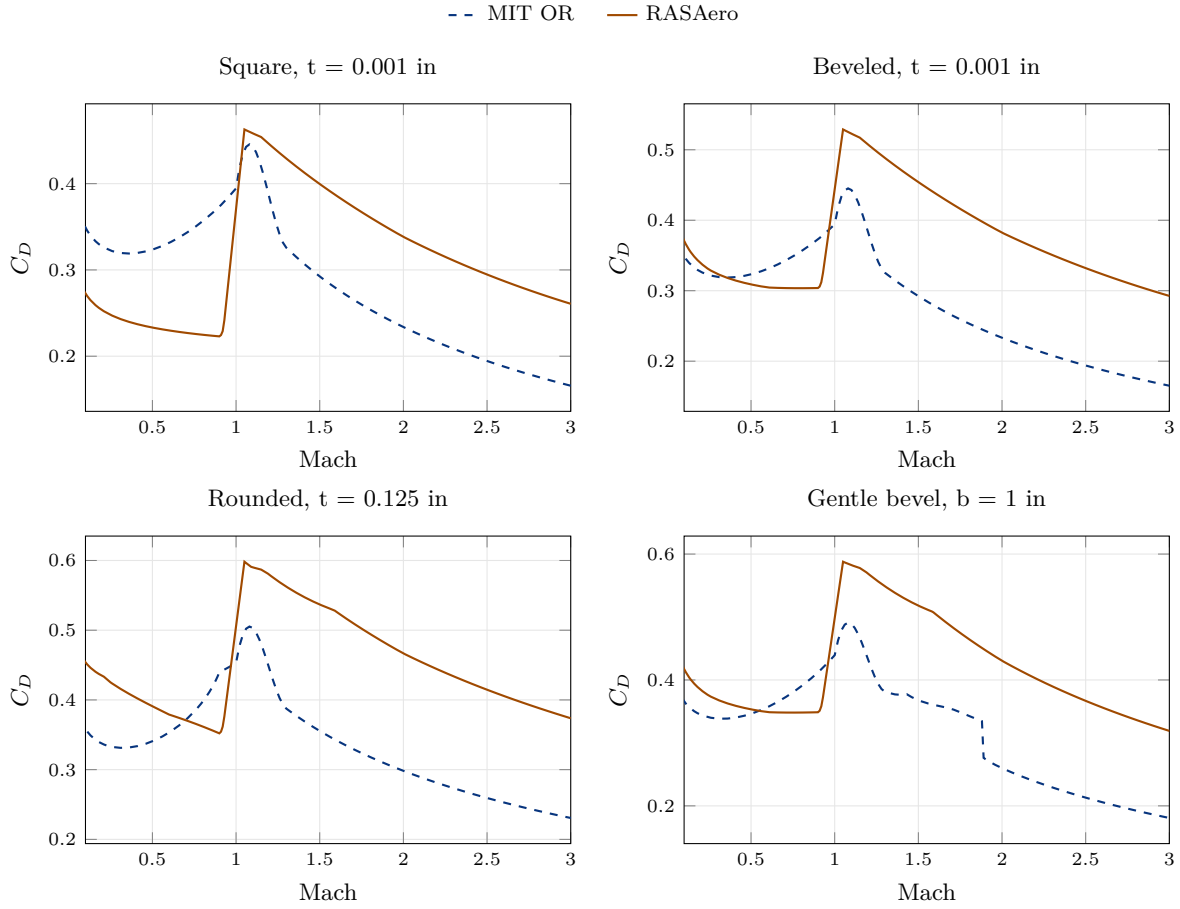


Figure 23. These limits test whether the models approach a consistent finite-planform, vanishing-thickness behavior.

Export files: 14-profile-limits.svg and 14-profile-limits.png.

Directory: aerodynamic-investigation/extended/figures/.

Geometric scale and Reynolds number

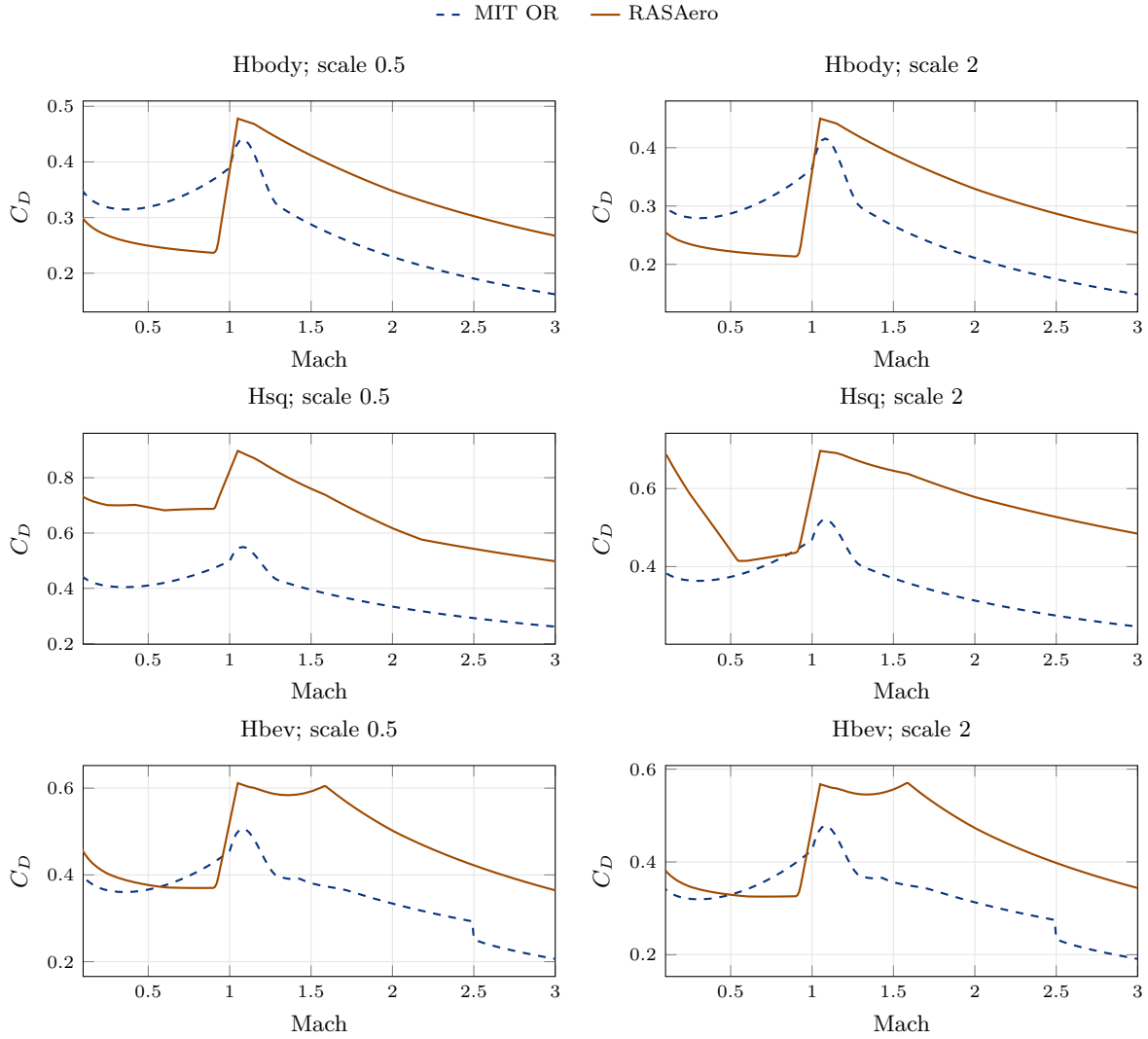


Figure 24. Every external length is scaled together; reference area follows diameter squared. At common Mach and atmosphere, dimensionless geometry is unchanged and Reynolds number varies.

Export files: 15-geometric-scale.svg and 15-geometric-scale.png.

Directory: aerodynamic-investigation/extended/figures/.

Fin drag versus upstream body length

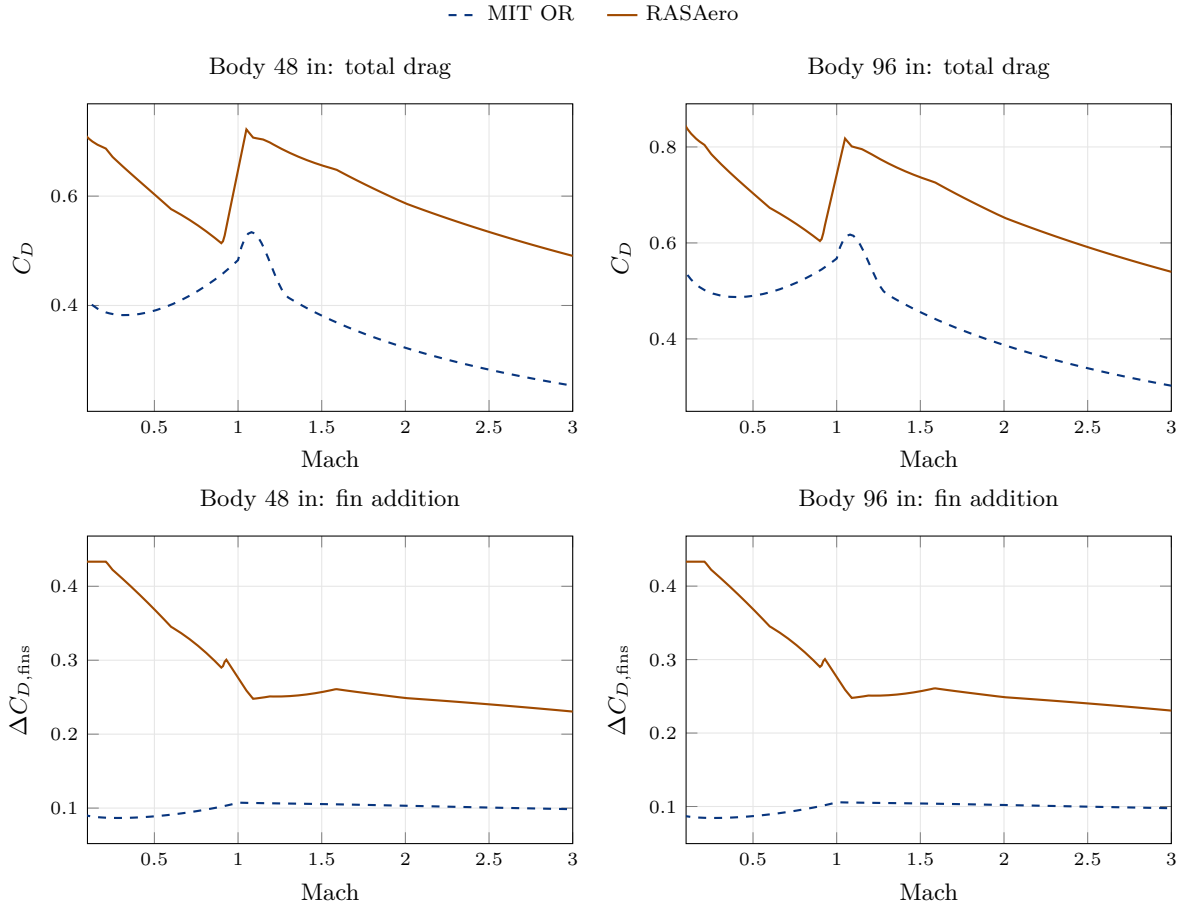


Figure 25. Only the cylindrical body length changes. Net fin additions expose the influence of the skin-friction reference length.

Export files: 16-fin-length-invariance.svg and 16-fin-length-invariance.png.

Directory: aerodynamic-investigation/extended/figures/.

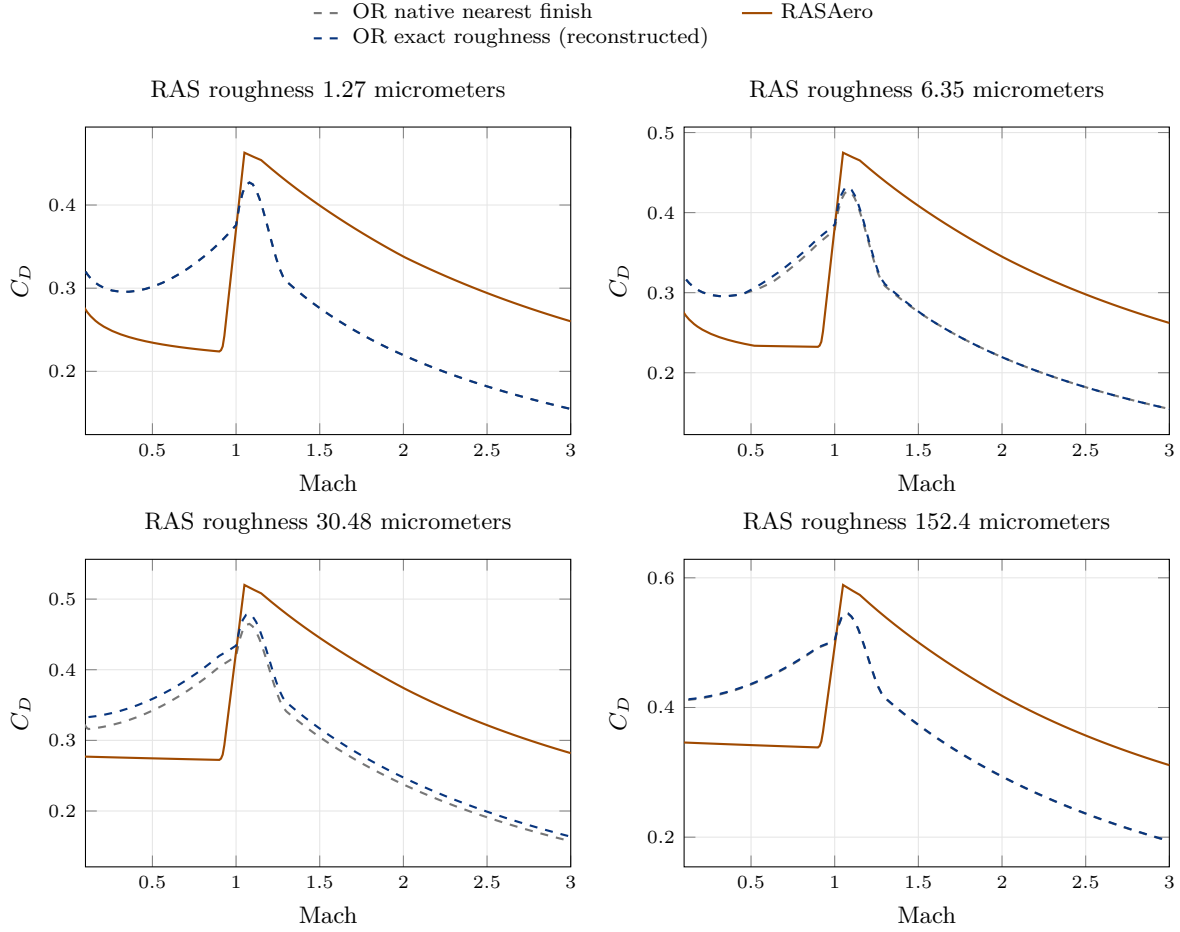
Physical roughness: Hbody

Figure 26. Gray: native OR nearest available finish (2, 5, 20, 150 micrometers respectively). Blue: native OR correlation reconstructed at the exact RAS roughness, verified against all native finish curves; no OR application changes.

Export files: 17-roughness-Hbody.svg and 17-roughness-Hbody.png.

Directory: aerodynamic-investigation/extended/figures/.

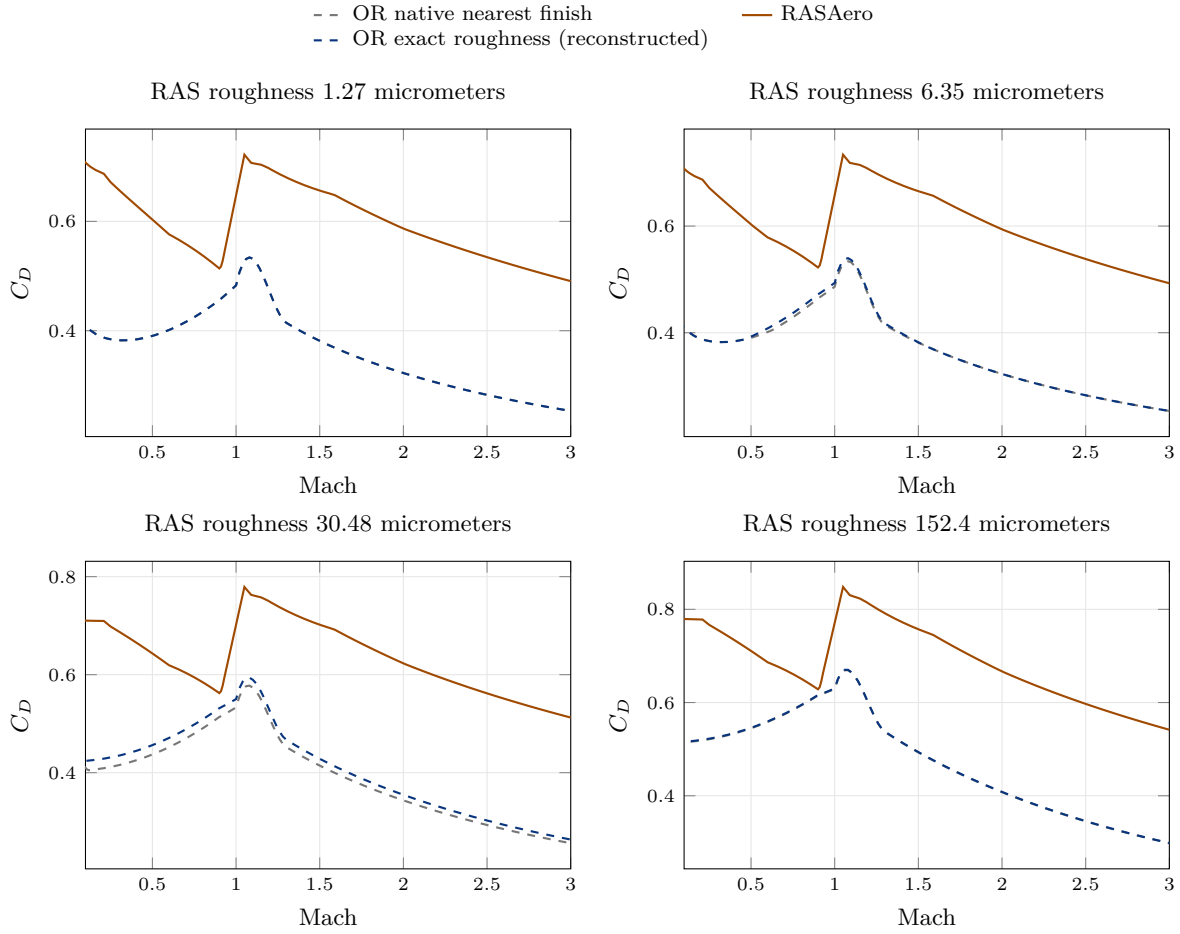
Physical roughness: Hsq

Figure 27. Gray: native OR nearest available finish (2, 5, 20, 150 micrometers respectively). Blue: native OR correlation reconstructed at the exact RAS roughness, verified against all native finish curves; no OR application changes.

Export files: 17-roughness-Hsq.svg and 17-roughness-Hsq.png.

Directory: aerodynamic-investigation/extended/figures/.

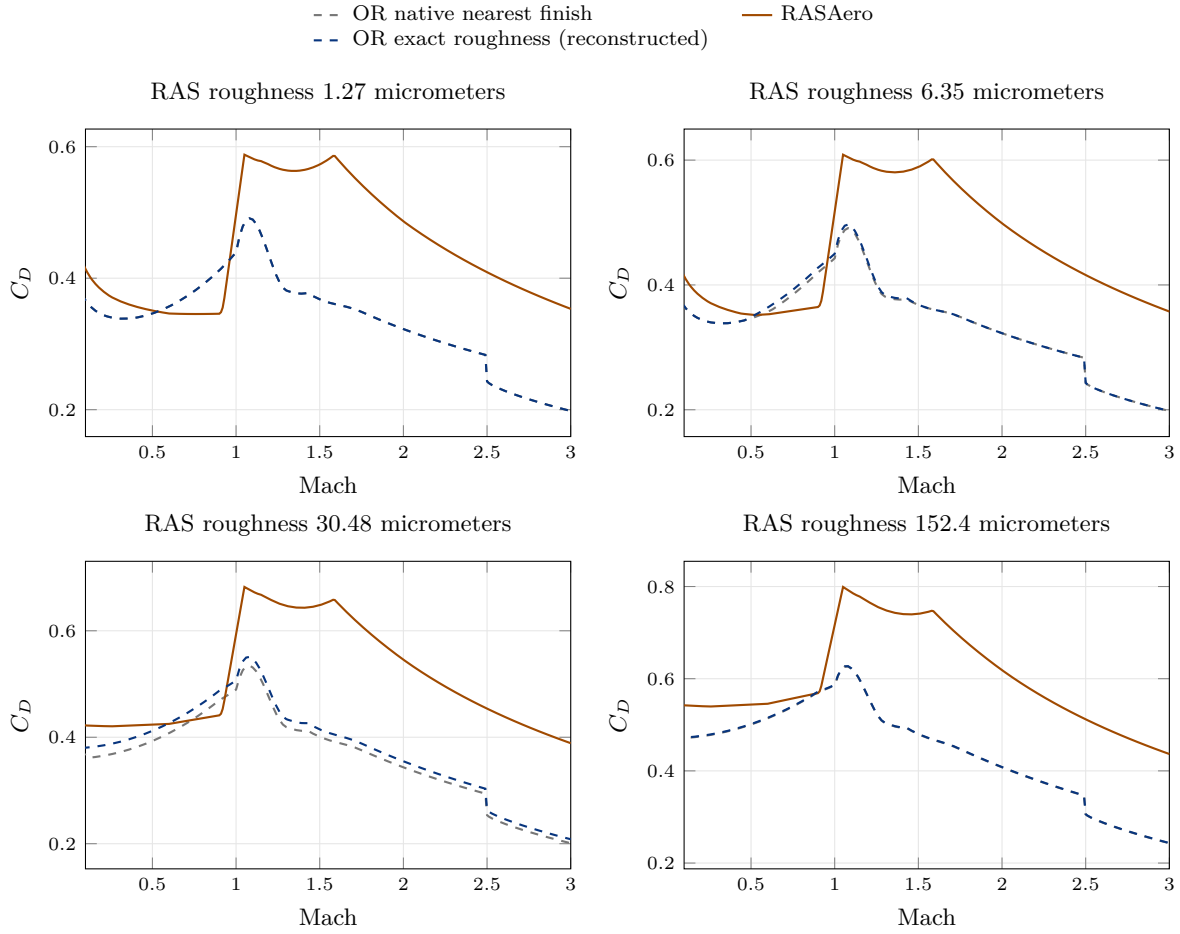
Physical roughness: Hbev

Figure 28. Gray: native OR nearest available finish (2, 5, 20, 150 micrometers respectively). Blue: native OR correlation reconstructed at the exact RAS roughness, verified against all native finish curves; no OR application changes.

Export files: 17-roughness-Hbev.svg and 17-roughness-Hbev.png.

Directory: aerodynamic-investigation/extended/figures/.

Roughness-induced drag at fixed Mach

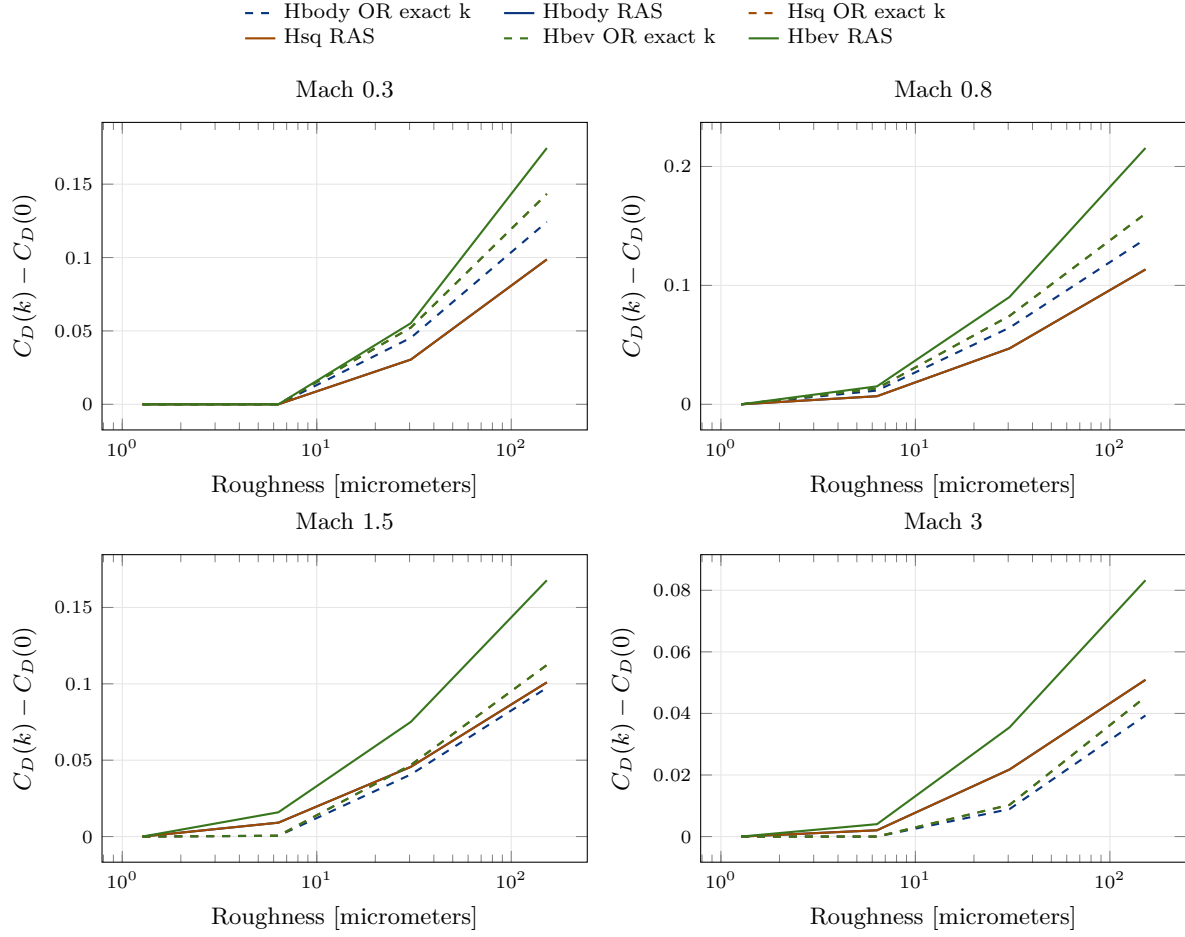


Figure 29. Numerical physical roughness is matched in the reconstructed OR correlation. Pressure and base terms retain their native OR values.

Export files: 18-roughness-increment.svg and 18-roughness-increment.png.

Directory: aerodynamic-investigation/extended/figures/.

Isolated fin response to physical roughness

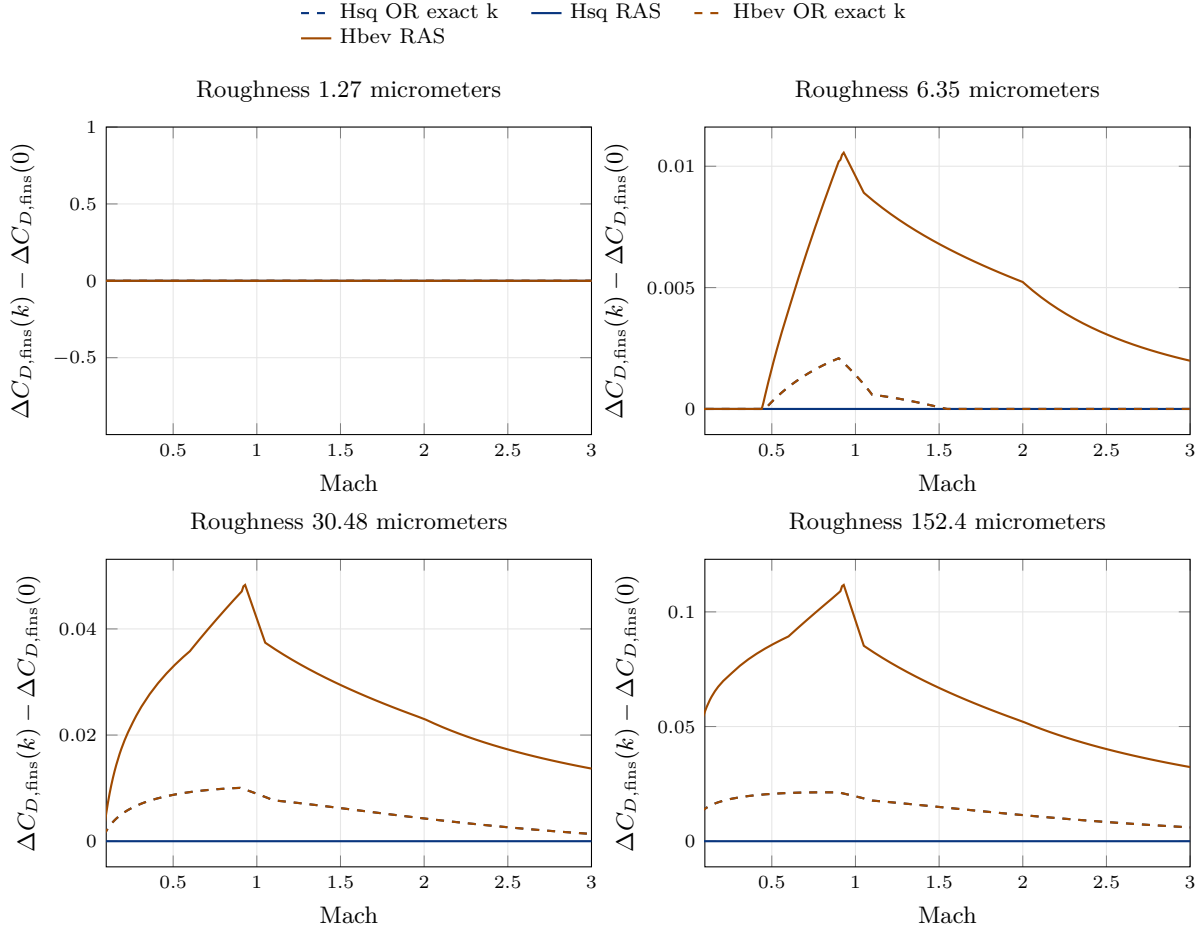


Figure 30. Two subtractions remove smooth baseline drag and the finless-body roughness response. RASAero square-fin curves remain at zero to floating-point precision; beveled fins respond. OR exact-roughness curves are reconstructed from its native correlation.

Export files: 27-fin-roughness-response.svg and 27-fin-roughness-response.png.

Directory: aerodynamic-investigation/extended/figures/.

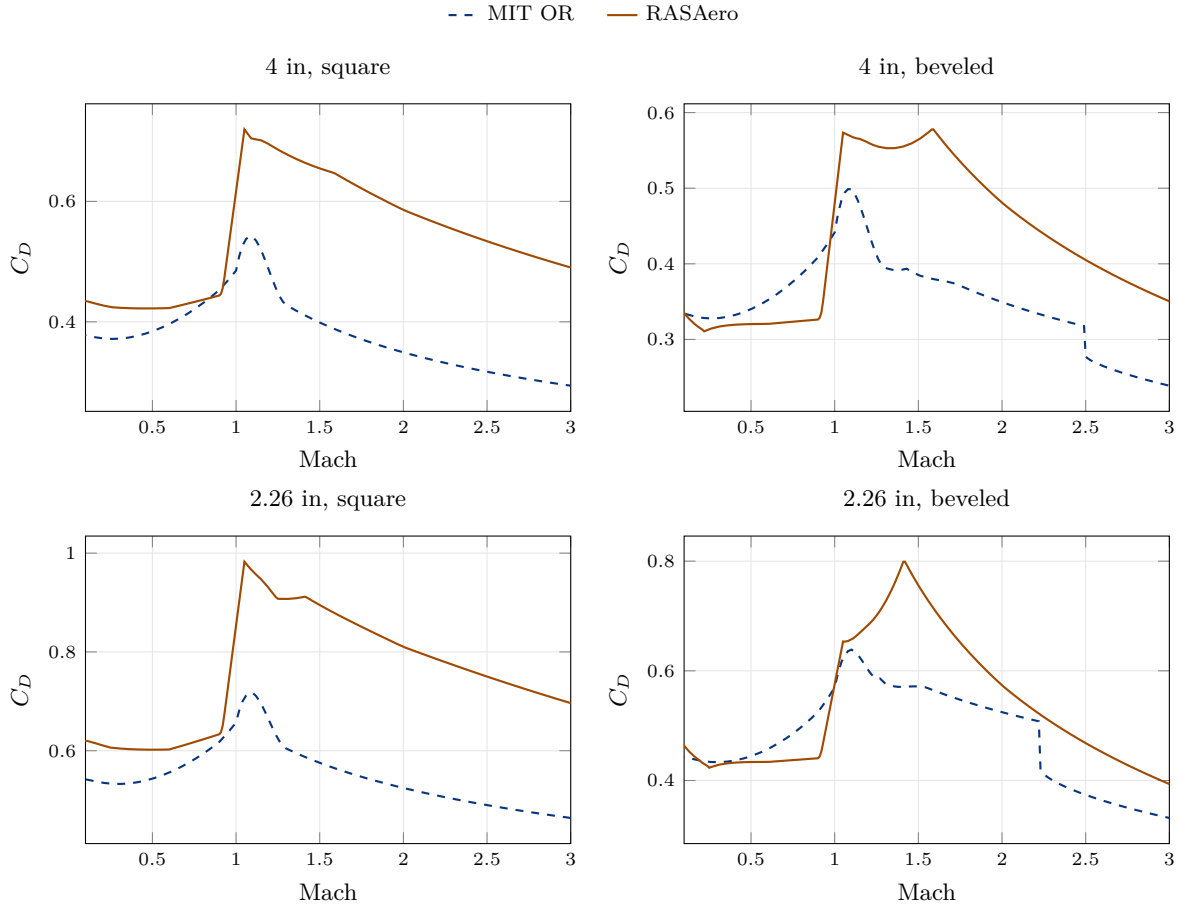
Laminar-to-turbulent option: finned rockets

Figure 31. OR perfectFinish=true and RAS All Turbulent Flow=false. These settings enable the respective transition models; their transition equations are not assumed identical.

Export files: 19-transition-finned.svg and 19-transition-finned.png.

Directory: aerodynamic-investigation/extended/figures/.

Sensitivity to flow-transition treatment

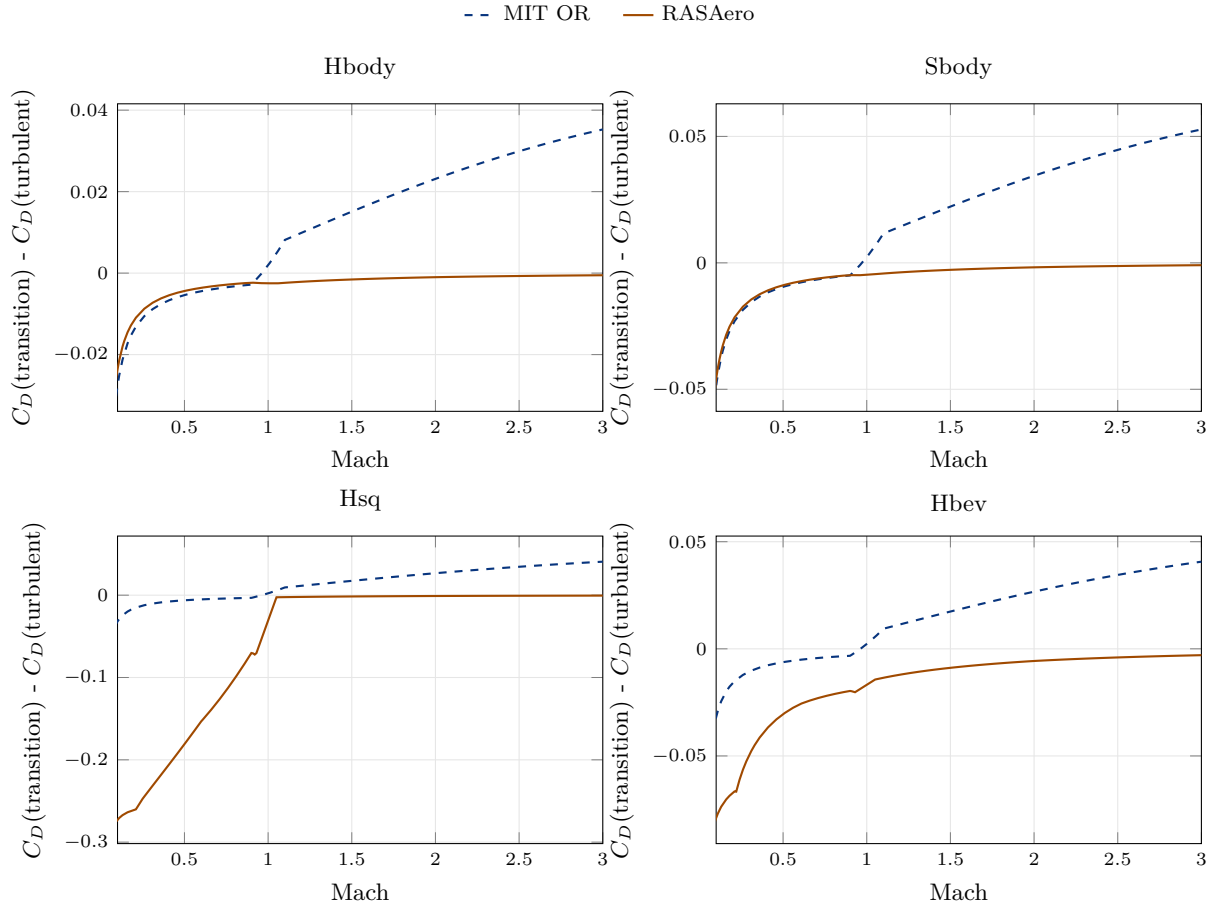


Figure 32. Subtracting the fully turbulent baseline separates the change due to each program's flow option from the baseline pressure-drag disagreement.

Export files: 20-transition-increment.svg and 20-transition-increment.png.

Directory: aerodynamic-investigation/extended/figures/.

Isolated fin response to flow transition

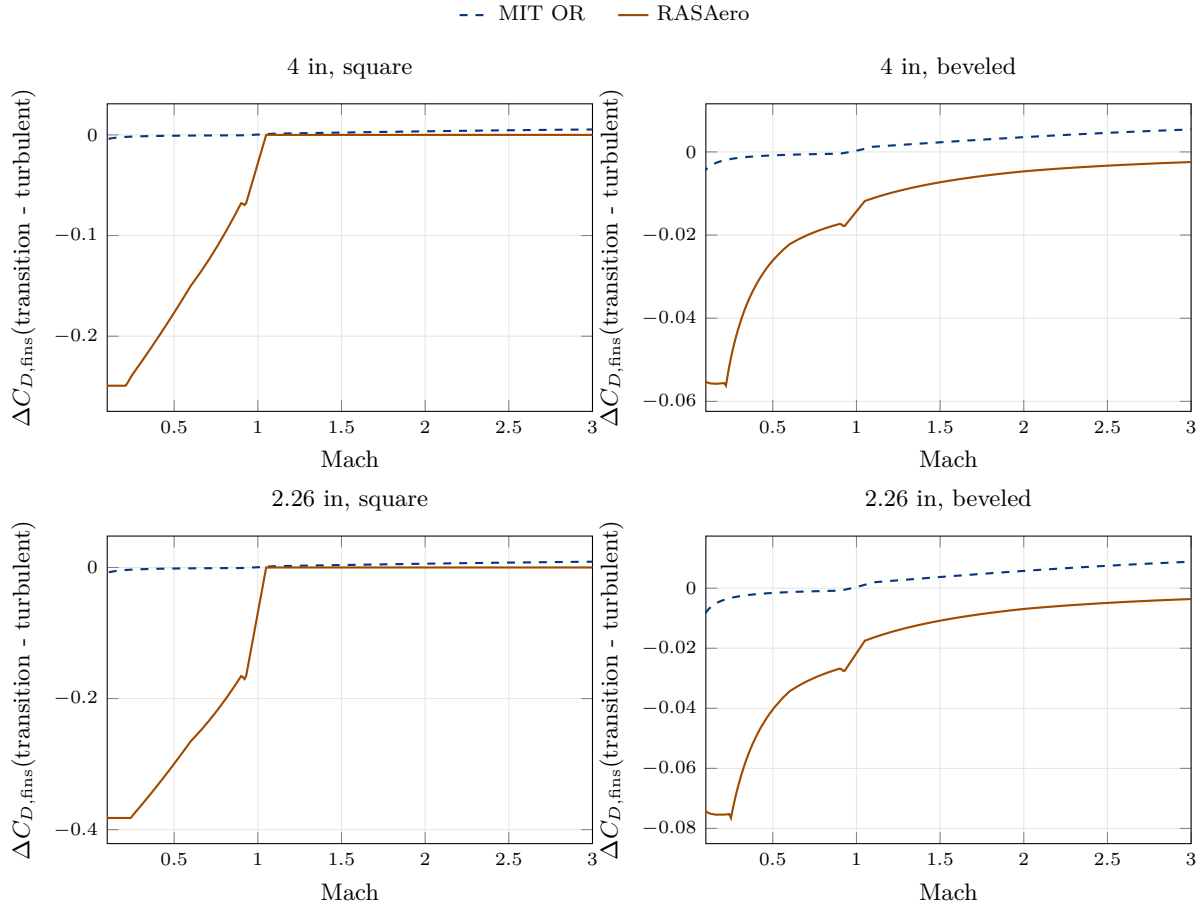


Figure 33. Subtract the body-only response to the flow option from the finned-rocket response. This isolates the net fin dependency without labeling it as a proprietary RASAero internal pressure or friction term.

Export files: 28-fin-transition-response.svg and 28-fin-transition-response.png.

Directory: aerodynamic-investigation/extended/figures/.

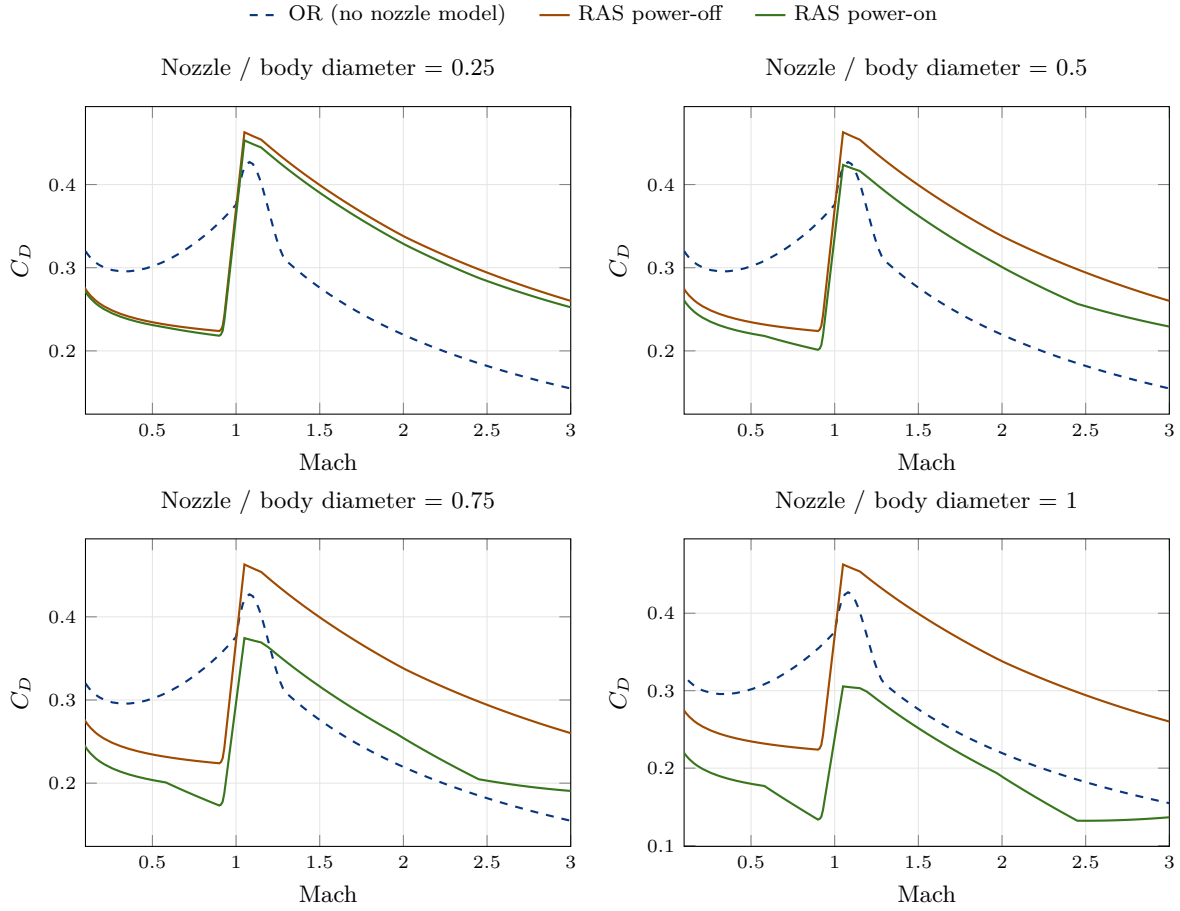
Nozzle-area sensitivity: Hbody

Figure 34. Aero-only probes. OR has no corresponding nozzle-area correction here; its baseline is shown to reveal the missing dependency, not as a matched powered-flow prediction.

Export files: 21-nozzle-Hbody.svg and 21-nozzle-Hbody.png.
 Directory: aerodynamic-investigation/extended/figures/.

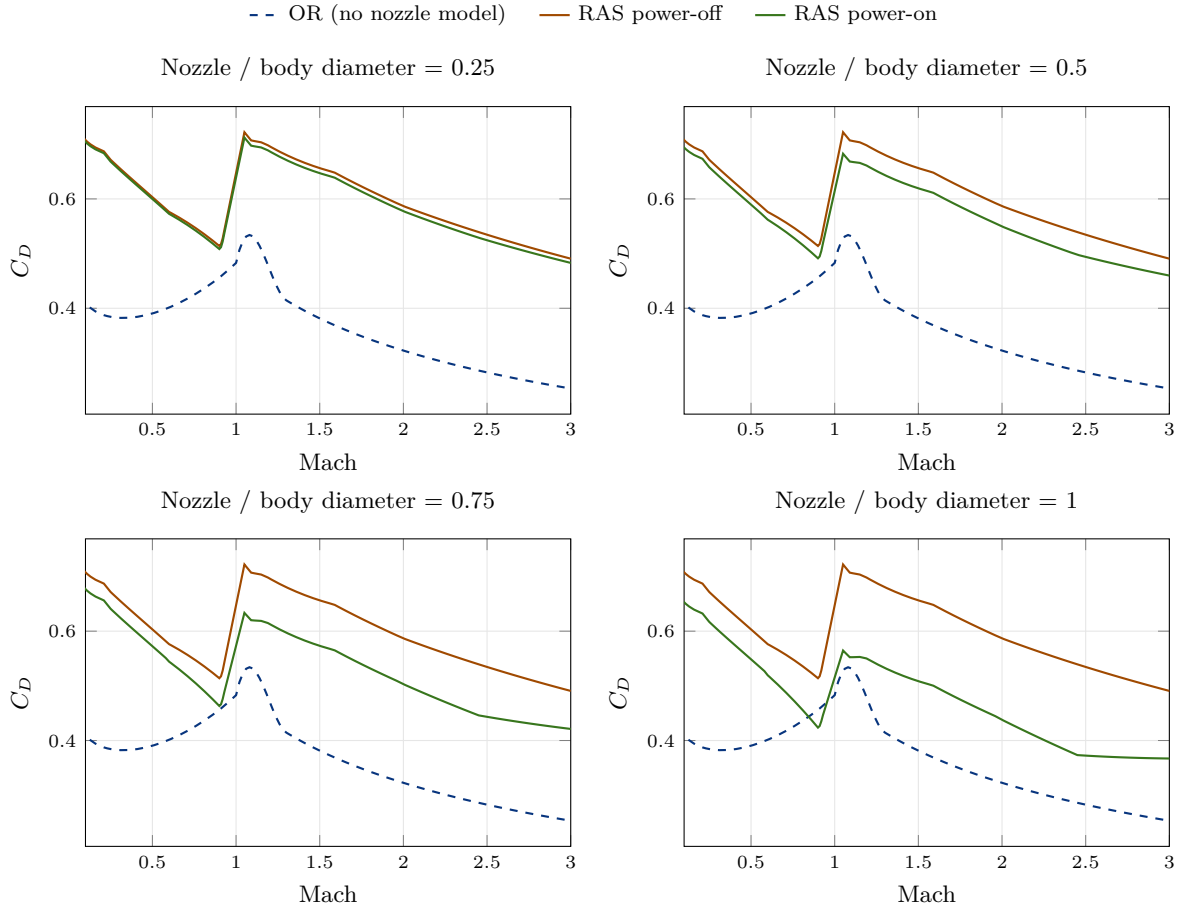
Nozzle-area sensitivity: Hsq

Figure 35. Aero-only probes. OR has no corresponding nozzle-area correction here; its baseline is shown to reveal the missing dependency, not as a matched powered-flow prediction.

Export files: 21-nozzle-Hsq.svg and 21-nozzle-Hsq.png.
 Directory: aerodynamic-investigation/extended/figures/.

Test of nozzle-area scaling

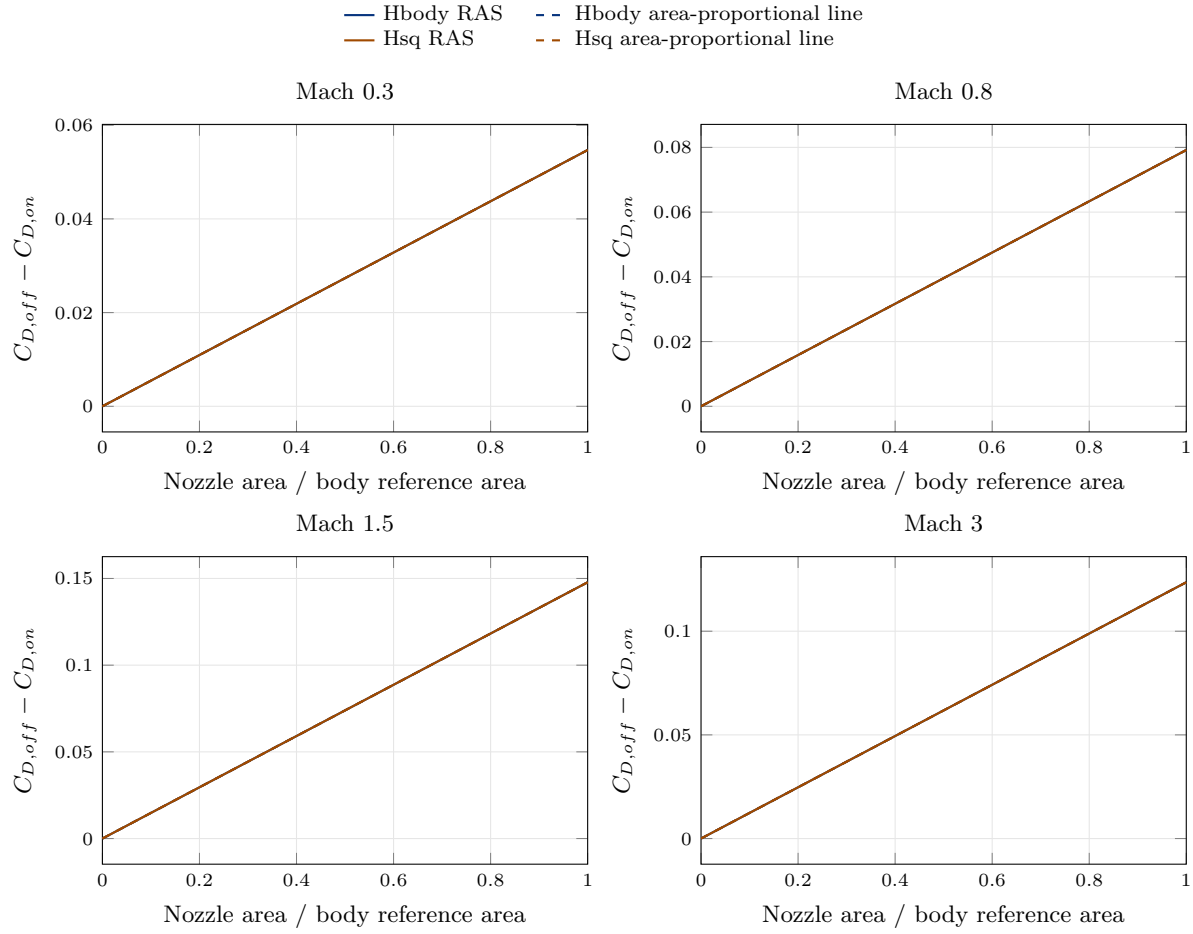


Figure 36. The dashed line uses the observed full-area value. It tests area proportionality; it does not identify the absolute body-base coefficient.

Export files: 22-nozzle-area-law.svg and 22-nozzle-area-law.png.

Directory: aerodynamic-investigation/extended/figures/.

Wind-axis drag at 0, 2 and 4 degrees

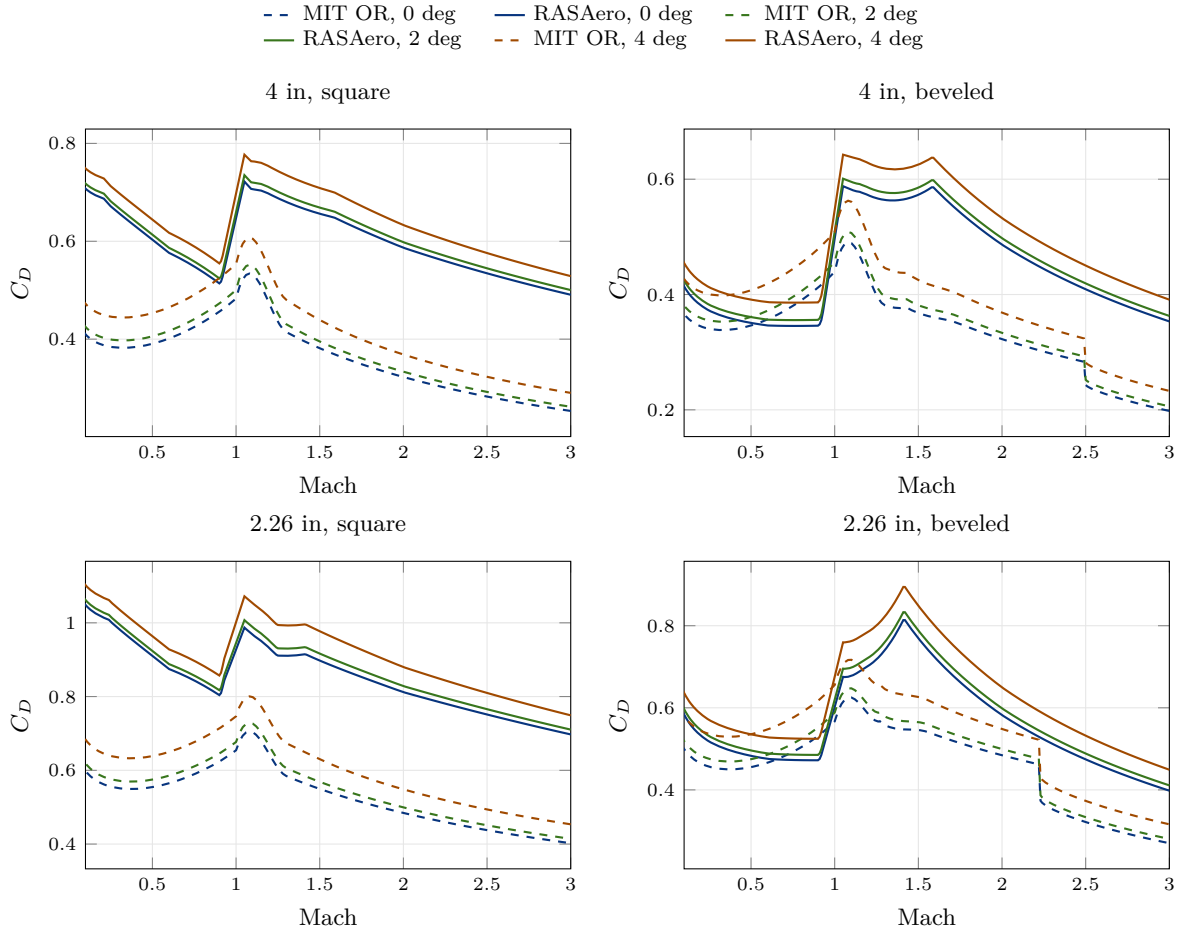


Figure 37. Body-axis forces are compared directly. Wind-axis drag/lift use $C_D = C_A \cos(\alpha) + C_N \sin(\alpha)$, $C_L = C_N \cos(\alpha) - C_A \sin(\alpha)$. RAS lift is reconstructed for power-off; its raw CL column corresponds to power-on when nozzle area is nonzero.

Export files: 23-alpha-CD.svg and 23-alpha-CD.png.

Directory: aerodynamic-investigation/extended/figures/.

Axial force at 0, 2 and 4 degrees

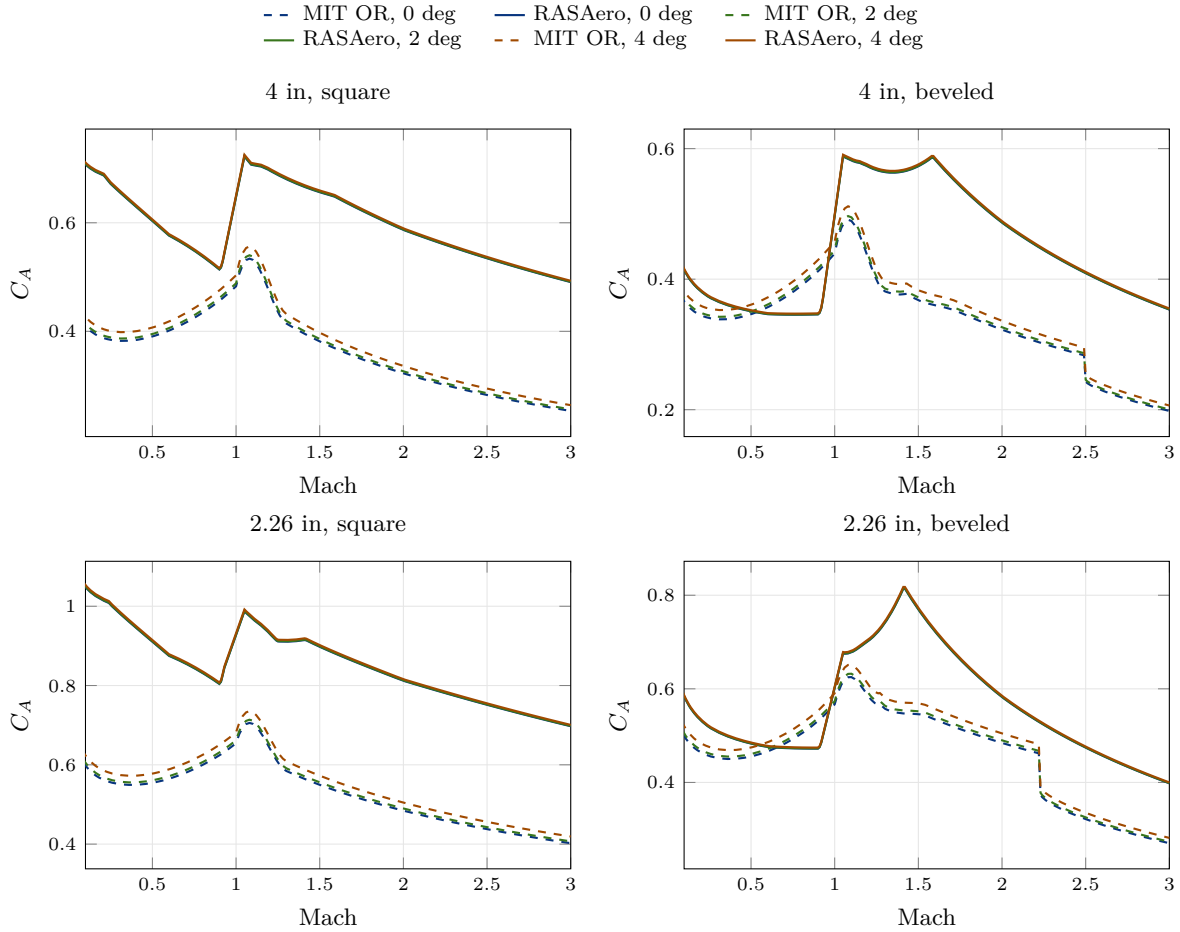


Figure 38. Body-axis forces are compared directly. Wind-axis drag/lift use $C_D = C_A \cos(\alpha) + C_N \sin(\alpha)$, $C_L = C_N \cos(\alpha) - C_A \sin(\alpha)$. RAS lift is reconstructed for power-off; its raw CL column corresponds to power-on when nozzle area is nonzero.

Export files: 23-alpha-CA.svg and 23-alpha-CA.png.

Directory: aerodynamic-investigation/extended/figures/.

Normal force at 0, 2 and 4 degrees

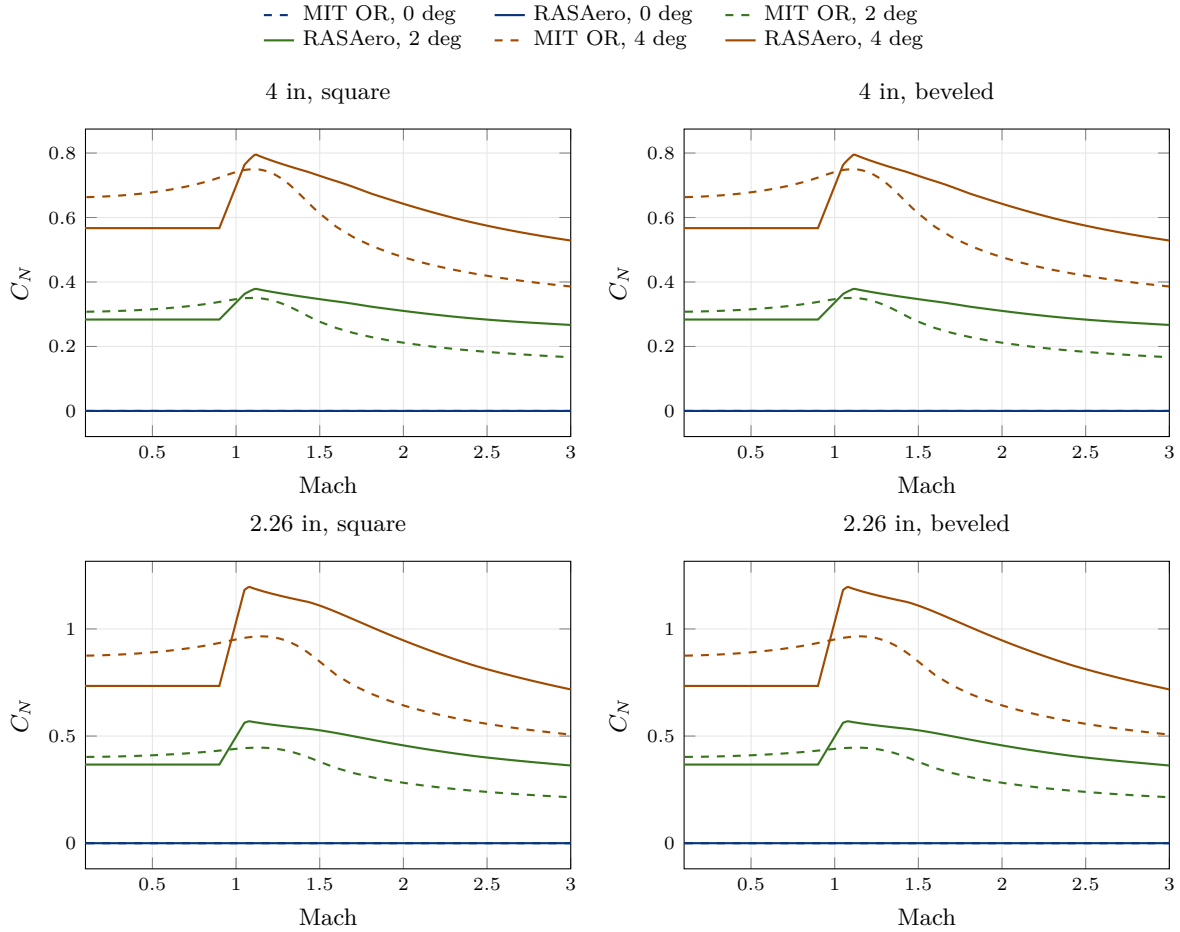


Figure 39. Body-axis forces are compared directly. Wind-axis drag/lift use $C_D = C_A \cos(\alpha) + C_N \sin(\alpha)$, $C_L = C_N \cos(\alpha) - C_A \sin(\alpha)$. RAS lift is reconstructed for power-off; its raw CL column corresponds to power-on when nozzle area is nonzero.

Export files: 23-alpha-CN.svg and 23-alpha-CN.png.

Directory: aerodynamic-investigation/extended/figures/.

Wind-axis lift at 0, 2 and 4 degrees

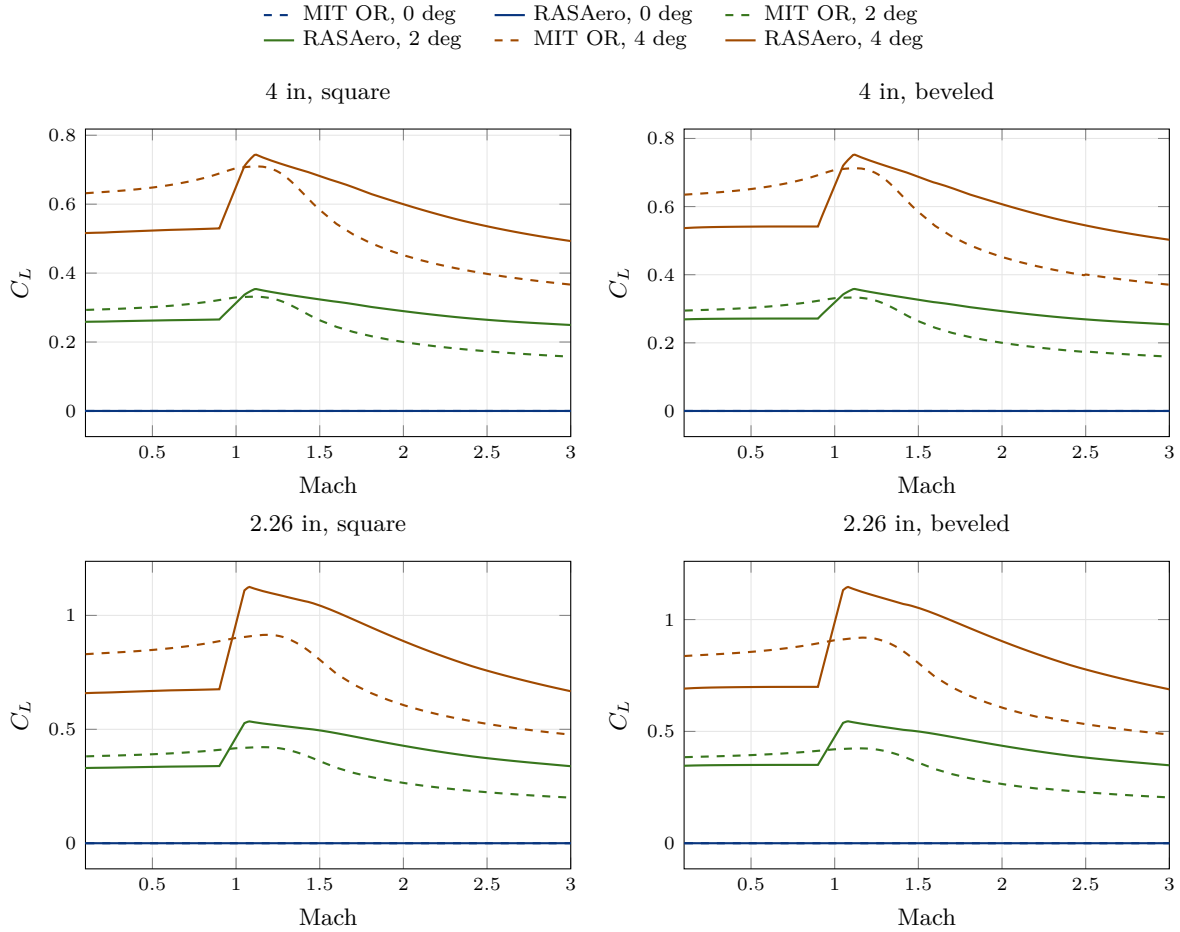


Figure 40. Body-axis forces are compared directly. Wind-axis drag/lift use $C_D = C_A \cos(\alpha) + C_N \sin(\alpha)$, $C_L = C_N \cos(\alpha) - C_A \sin(\alpha)$. RAS lift is reconstructed for power-off; its raw CL column corresponds to power-on when nozzle area is nonzero.

Export files: 23-alpha-CL.svg and 23-alpha-CL.png.

Directory: aerodynamic-investigation/extended/figures/.

Center of pressure at 0 degrees

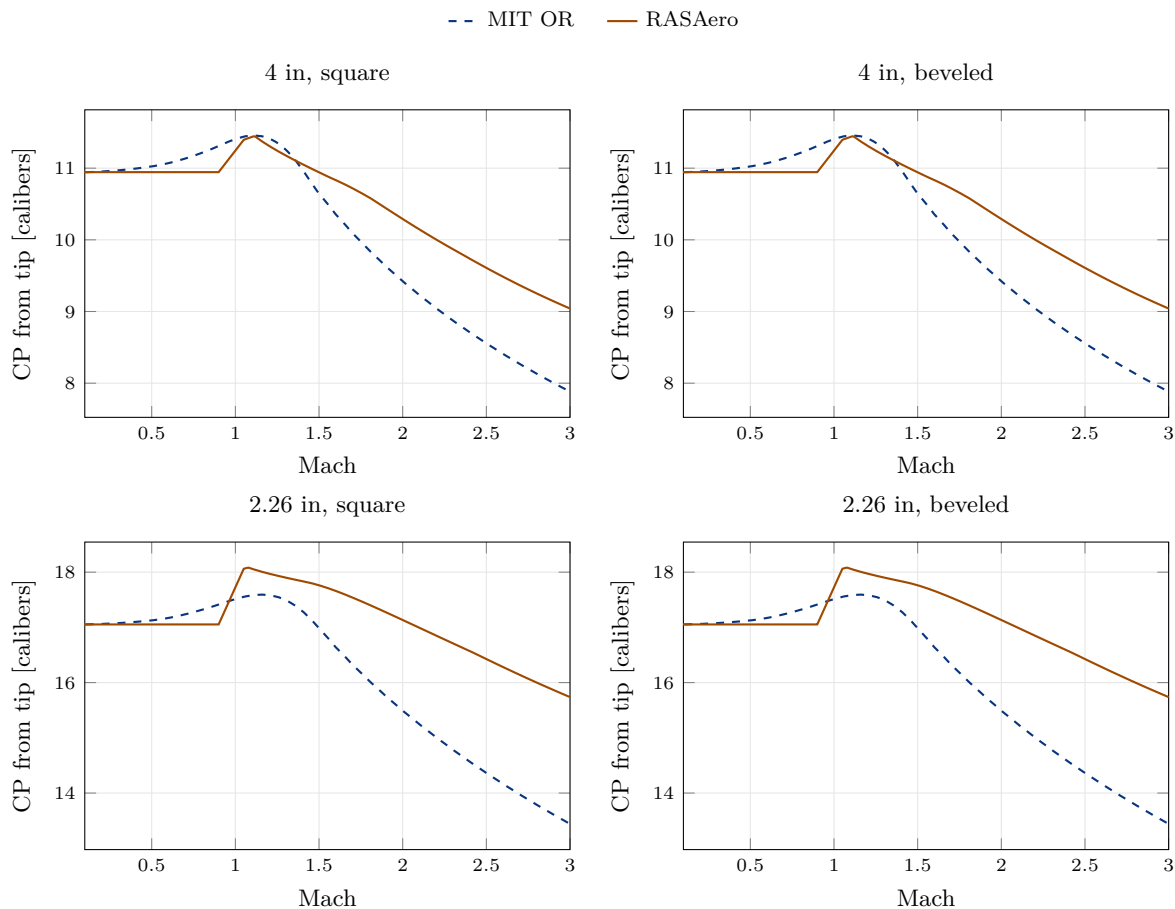


Figure 41. CP is measured from the nose tip in body diameters; this is not a stability margin. OR warns of limitations of its body-force treatment above Mach 1.1.

Export files: 24-cp-0.svg and 24-cp-0.png.

Directory: aerodynamic-investigation/extended/figures/.

Center of pressure at 4 degrees

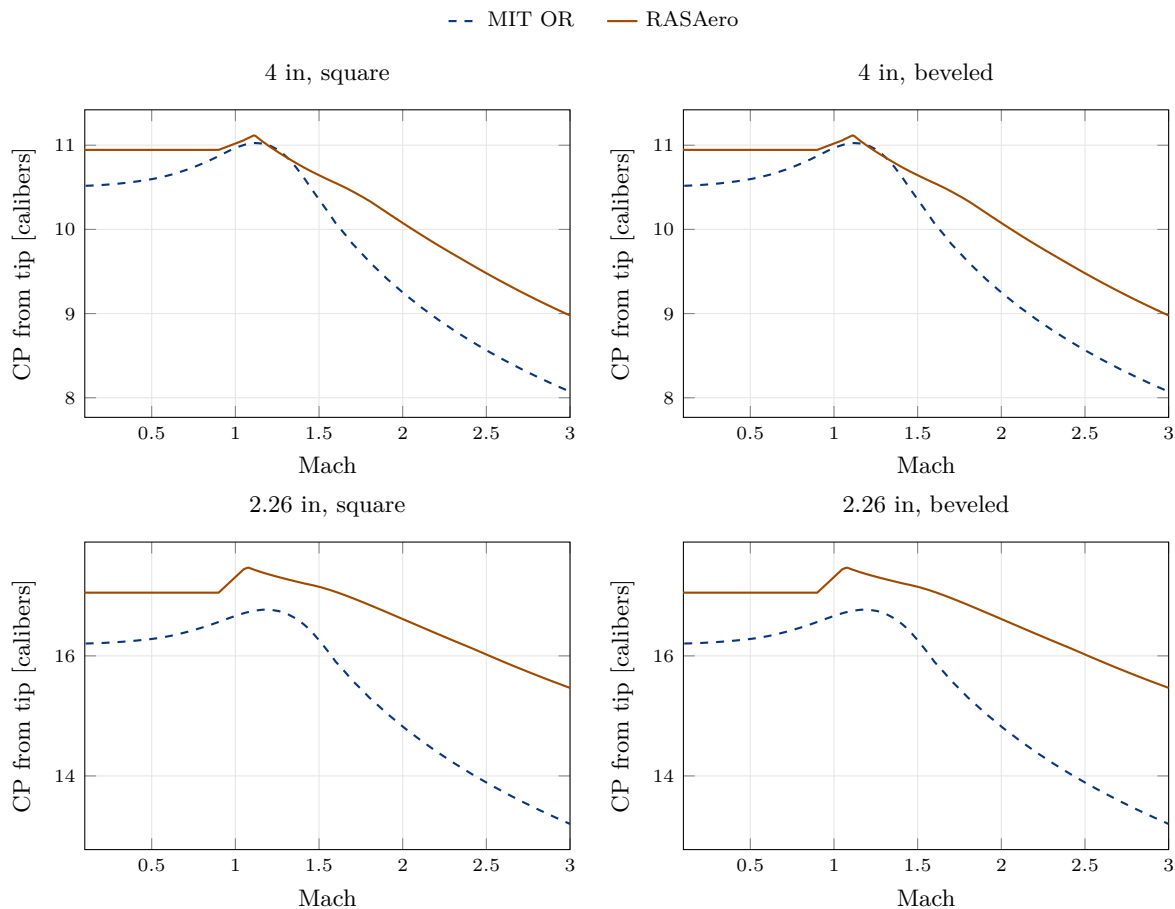


Figure 42. CP is measured from the nose tip in body diameters; this is not a stability margin. OR warns of limitations of its body-force treatment above Mach 1.1.

Export files: 24-cp-4.svg and 24-cp-4.png.

Directory: aerodynamic-investigation/extended/figures/.

Normal-force slope over 0 to 4 degrees

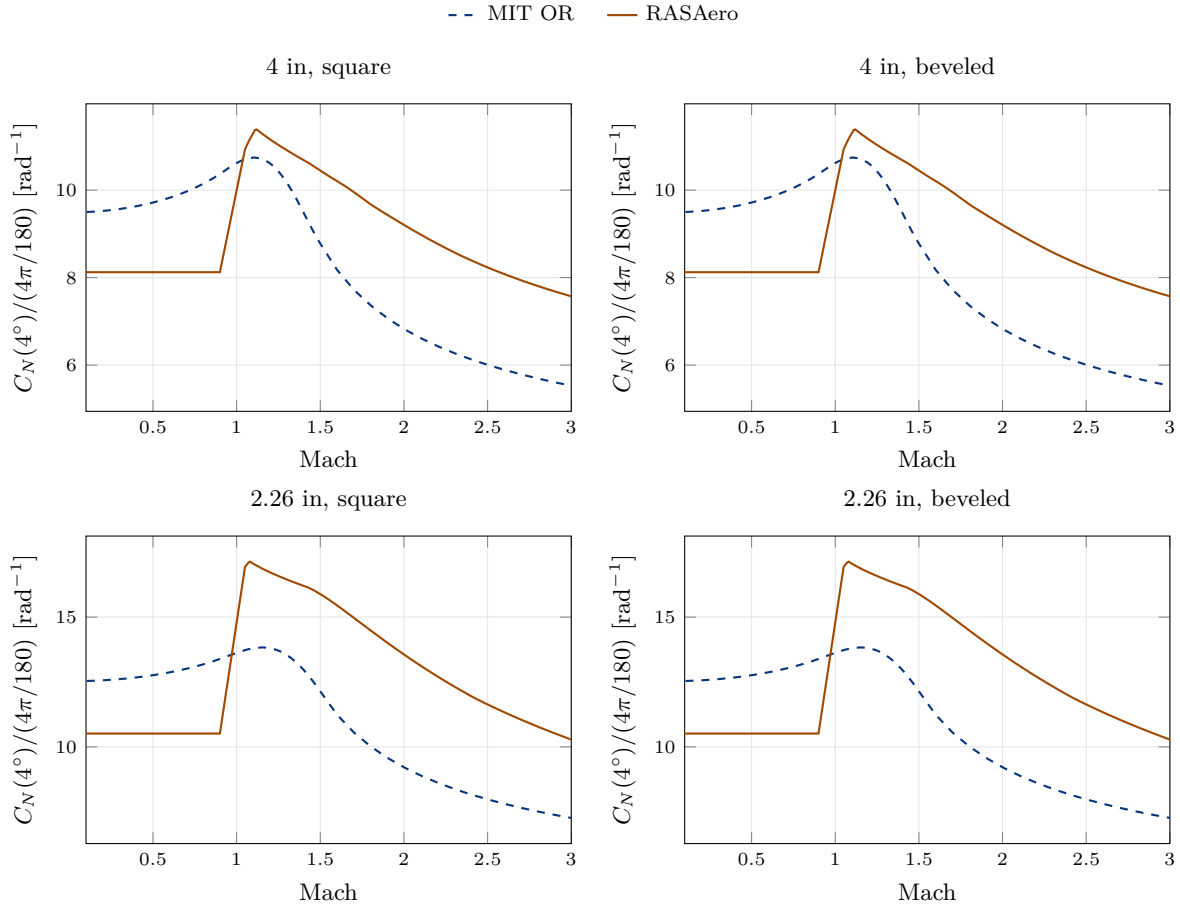


Figure 43. Both plotted slopes use the same 0-to-4-degree secant definition; neither is assumed to be the infinitesimal derivative in nonlinear flow.

Export files: 25-cna-secant.svg and 25-cna-secant.png.

Directory: aerodynamic-investigation/extended/figures/.

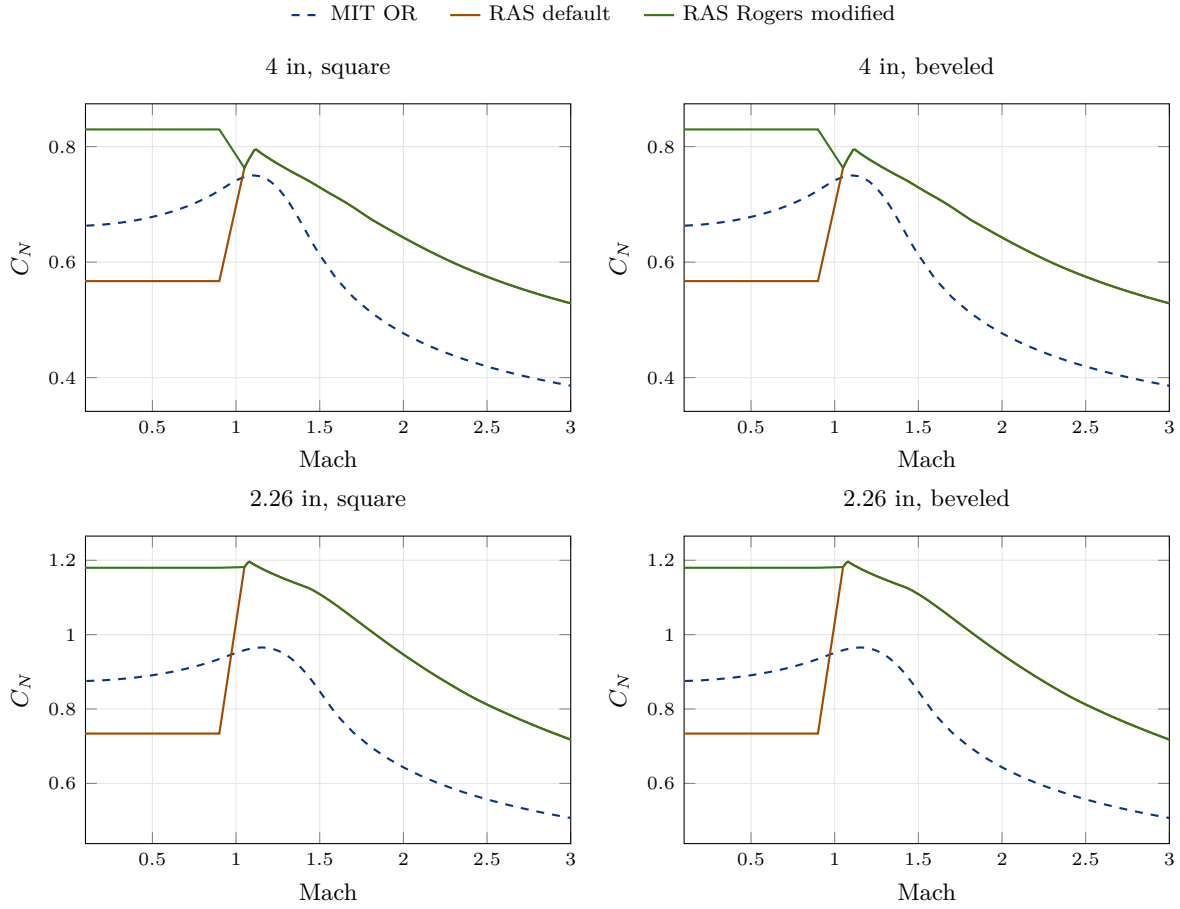
Rogers Modified Barrowman: CN at 4 degrees

Figure 44. Default and modified RAS settings share geometry and drag-flow inputs. This isolates the normal-force/body-interference option, compared with unmodified MIT OR.

Export files: 26-rogers-CN.svg and 26-rogers-CN.png.

Directory: aerodynamic-investigation/extended/figures/.

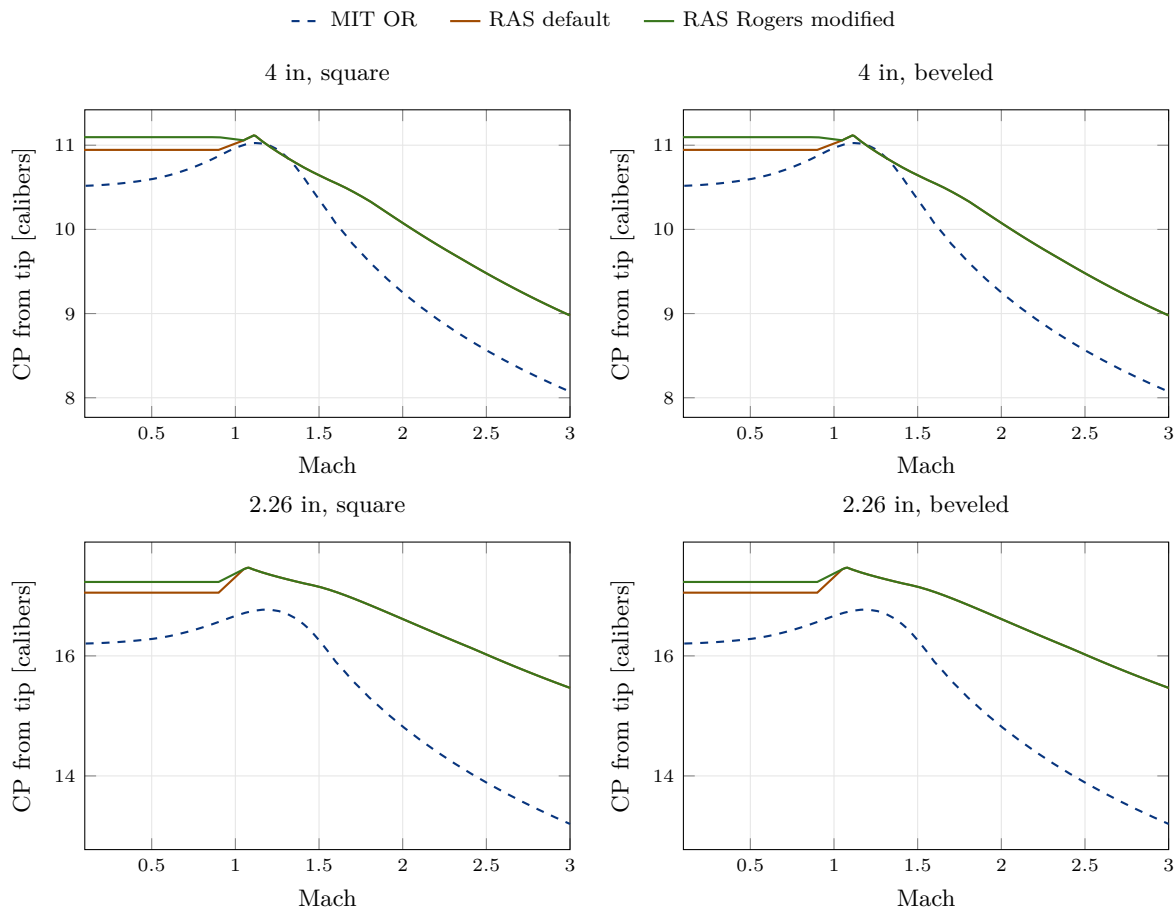
Rogers Modified Barrowman: CP at 4 degrees

Figure 45. Default and modified RAS settings share geometry and drag-flow inputs. This isolates the normal-force/body-interference option, compared with unmodified MIT OR.

Export files: 26-rogers-CP.svg and 26-rogers-CP.png.

Directory: aerodynamic-investigation/extended/figures/.

Rogers Modified Barrowman: CD at 4 degrees

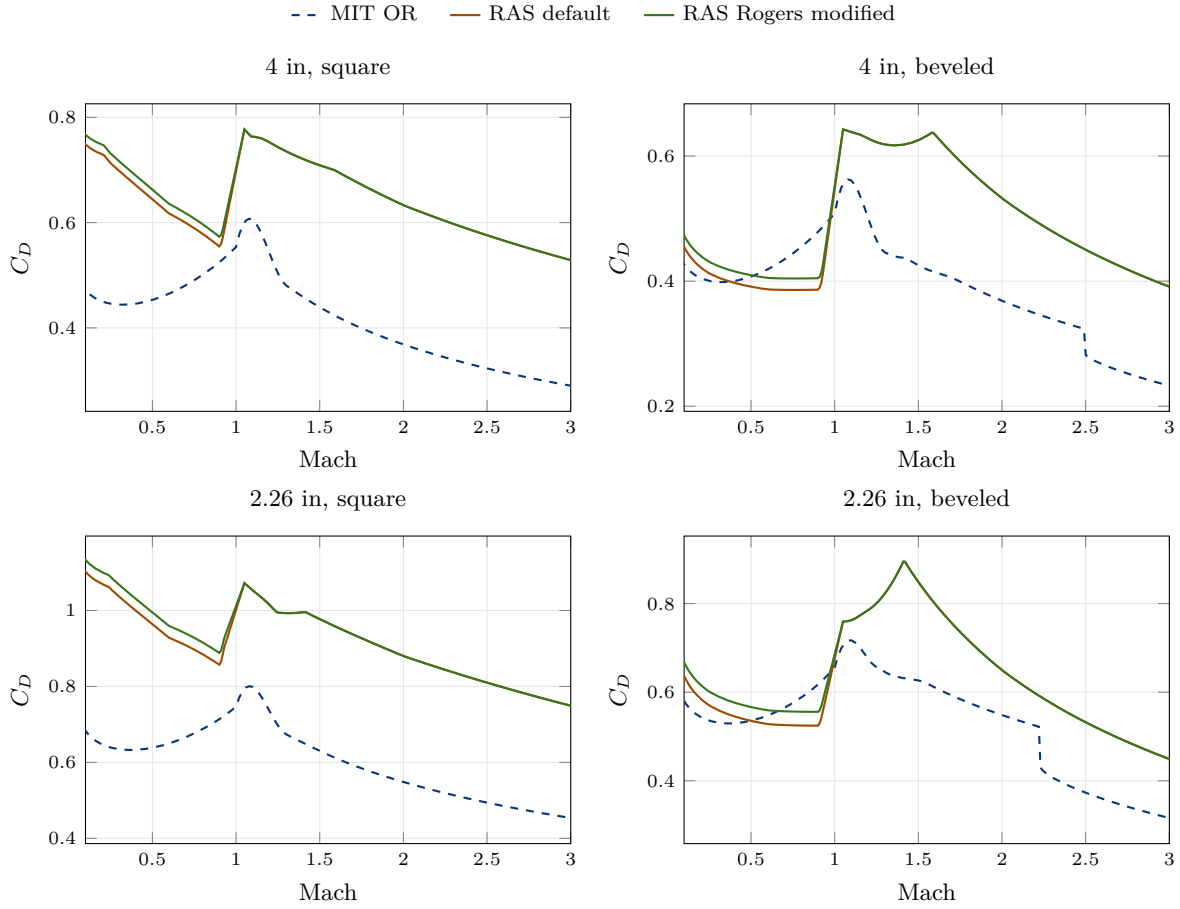


Figure 46. Default and modified RAS settings share geometry and drag-flow inputs. This isolates the normal-force/body-interference option, compared with unmodified MIT OR.

Export files: 26-rogers-CD.svg and 26-rogers-CD.png.

Directory: aerodynamic-investigation/extended/figures/.